FRC 2023 CHARGED UP -- Robot Design Deep Dive

Design reference for CHARGED UP. Each profile comes from the team's own public material: CAD (Onshape, GrabCAD, Fusion 360, STEP), technical binders, code, and Chief Delphi release and reveal threads. Performance data comes from The Blue Alliance.

The CHARGED UP design landscape

The game

CHARGED UP is a cone and cube game. Robots collect CONES and CUBES from the floor or from the SINGLE and DOUBLE SUBSTATIONS. Robots score them on the three rows of the GRID (hybrid, mid, high) to make LINKS. In the endgame, robots DOCK and ENGAGE on the CHARGE STATION. Many designs used an arm, a telescoping arm or an elevator with an end effector for cones and cubes. Auto-balance on the CHARGE STATION was part of most top autonomous routines.

How to use this document

The first section lists the profiled robots in the top 30 by Local EPA. The second section lists the other profiled robots. Each profile gives the archetype, the subsystems, vision, endgame, autonomous routines, the public CAD or binder, and lessons for students. A star marks a study pick. Each study pick has downloadable CAD or a full technical binder and one specific design lesson. "Unconfirmed" means the public material does not state the detail. It does not mean the robot does not have it.

About the EPA numbers

The Statbotics data API is offline. EPA is Local EPA (approx): it was computed from The Blue Alliance match data with the open-source Statbotics model (github.com/avgupta456/statbotics). The rank order follows Statbotics. The absolute values can differ from statbotics.io. Records are win-loss-tie totals from official events on The Blue Alliance.

Profiled robots at a glance (sorted by Local EPA)

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**5940 BREAD**	Pain d' Epi	**74**	#3	40-9	Pain d' Epi: tilted Dyneema elevator; cube/cone floor intake whose spr	Onshape CAD
	6329 The Bucks' Wrath	Shock	**74**	#2	55-14	'Shock': two-position tipping elevator; dual-motor intake with a wrist	CD thread
■	**2056 OP Robotics**	UPPERCUT	**73**	#4	61-8	'UPPERCUT': MK4i swerve; arm with gas-shock shoulder hold; driver-set	Tech binder
■	**2910 Jack in the Bot**	Phantom	**73**	#6	69-16	'Phantom': 26x28 robot under 17 in stowed, 2-stage cascade-rigged tele	Onshape CAD
■	**3005 RoboChargers**		**73**	#5	58-12	Tower-mounted chain-driven arm with a pivoting, turnbuckle-tensioned g	Onshape CAD
	4414 HighTide	Tsunami	**71**	#8	60-10-1	'Tsunami': slanted elevator with a v4b and intake; before Chezy Champs	CD thread
■	**111 WildStang**		**70**	#16	38-5-1	Lift, belt-driven arm and wrist with a cone-wedging gripper; double su	Onshape CAD
■	**2075 Enigma Robotics**	Cypher	**69**	#18	55-15	'Cypher': arm with shoulder and elbow joints; ground cube intake; Lime	Onshape CAD
	1577 Steampunk	Jesko	**68**	#19	51-12	'Jesko': arm with telescoping extension; ground intake for tipped cone	CD thread
■	**6036 Peninsula Robotics**	Dum-E	**68**	#21	49-18-1	'Dum-E': arm and ground intake with chain drives; cameras moved to the	GrabCAD
	125 NUTRONs		**66**	#23	62-25-3	Multi-stage telescoping arm (52 in of extension per a team reply) with	CD thread
■	**254 The Cheesy Poofs**	BREAKDOWN	**66**	#28	36-5	'BREAKDOWN': rear vertical elevator, tilted 3-stage 'Laterator', one-s	Tech binder
■	**2767 Stryke Force**	Snake Eyes	**66**	#29	52-14	'Snake Eyes': carbon fiber elevator pivoted by printed sector gears, a	Onshape CAD
	3467 Windham Windup		**66**	#25	55-17-4	Double-jointed arm (per interview summary) with pneumatic claw and wri	CD thread
	7157 µBotics	Tai Lung	**65**	#33	33-11-1	'Tai Lung': 65 deg elevator with an arm (one Falcon, 80:1); fork buddy	Onshape CAD
■	**5460 Strike Zone**	VIPER	**64**	#44	64-28	'VIPER': SDS MK4i swerve; Falcon intake that hands off cones; 2-Falcon	Onshape CAD
■	**6672 Fusion Corps**	TARS	**62**	#64	55-19	'TARS': swerve; pivoting arm with forearm and wrist gearboxes; live-ro	Onshape CAD
■	**3357 COMETS**	Kushida	**61**	#79	63-20	'Kushida': 21x26 in chain-in-tube tank drive; 3-stage elevator with a	GrabCAD
■	**341 Miss Daisy**	Miss Daisy XXI	**59**	#105	42-19-1	'Miss Daisy XXI': belt-driven telescoping arm; roller-claw intake on a	Onshape CAD
	695 Bison Robotics	Goldmodule	**59**	#109	52-11	'Goldmodule': cascading elevator with an Everybot intake (per Behind t	Drive folder / doc
■	**1706 Ratchet Rockers**	Whiplash	**59**	#111	38-11-1	'Whiplash': cone and cube shooter on swerve; frame corners cut to the	Onshape CAD
■	**4795 Eastbots**	HYDRA	**59**	#106	50-15-1	'HYDRA': mid-level scorer; roller claw on a wrist; roller bar cube int	Onshape CAD
■	**8177 Vector**	Velocity	**58**	#112	36-24-2	'Velocity': MK4i L3 swerve, 55 deg tilted elevator, NEO wrist, triple-	Onshape CAD
■	**2337 EngiNERDs**	Vision	**57**	#145	45-17-1	'Vision': shoulder, elbow and wrist arm with rope-driven elbow and cab	Onshape CAD
	177 Bobcat Robotics	The Bobcat	**54**	#191	48-19	'The Bobcat': double-jointed arm on an angled elevator; swerve (the te	Onshape CAD
■	**3847 Spectrum**	X-Ray 2023	**54**	#182	44-20	'X-Ray 2023': cube-shooting swerve robot (per Behind the Bumpers) with	CAD release
■	**955 Ctrl C**		**53**	#212	44-28	Roller claw on an extension that cannot reach the floor; ground cube i	Onshape CAD
■	**2881 Lady Cans**	SCoRpion	**53**	#222	34-26	Two-lead-screw pivoting and extending arm with a suction-cup end effec	GrabCAD

	Team	Robot	EPA	Rank	Record	Archetype	Public design
■	**401 Copperhead Robotics**	Mushussu	**52**	#251	42-21-1	Double-sided arm (250-deg rotation, 30 in extension) with a 330-deg be	Onshape CAD
■	**7461 Sushi Squad**		**52**	#256	36-25	Chain-driven pivot arm with pneumatic extension and a claw; intake wit	Onshape CAD
	2877 LigerBots		**51**	#282	25-24	Swerve; pivoting 2-stage telescoping arm with cascade rigging and cons	Onshape CAD
■	**4201 The Vitruvian Bots**	Gridlock	**51**	#279	26-14	'Gridlock': 3-stage rope-driven elevator; 180-deg virtual 4-bar with 3	Onshape CAD
■	**1561 Roboducks**	Duck N' Cover V2	**50**	#327	39-20	Duck N' Cover V2: cube-only robot; pivoting arm with a cube shooter/in	CAD download
	4561 TerrorBytes	Fulcrum	**48**	#356	45-16-2	'Fulcrum': double-jointed arm on MAX Tube; Everybot-inspired intake ad	Onshape CAD
■	**131 CHAOS**		**42**	#560	20-25	Extender on linear sliders with a wrist and a light roller gripper; sw	CD thread
	4322 Clockwork		**42**	#571	16-9	Swerve on SwerveX modules with Falcon 500 motors; floor pickup limited	Onshape CAD
	3136 O.R.C.A.		**41**	#600	25-21	Tilted 3-stage elevator with intake on the middle carriage; narrow rec	Onshape CAD
	1018 Pike RoboDevils		**39**	#687	11-18	AM14U5 8-wheel kit chassis with NEOs; single-joint 'Sushi arm' planned	Onshape CAD
	3739 Oakbotics	Blue Steel	**39**	#697	20-28	'Blue Steel': MAXSwerve chassis; intake redesigned after week 1 (relea	GrabCAD
■	**3926 MPArors**	Snowdrift	**38**	#734	14-17	'Snowdrift': low-goal cycler on a slim modified KoP tank base with EVO	Onshape CAD
	1710 The Ravonics Revolution	SlipSTream	**37**	#821	27-20	'SlipSTream': double-jointed arm on 6 NEOs, pneumatic claw with self-l	Fusion 360 CAD
	8738 Slice		**35**	#905	28-23	Elevator and arm robot on a Kit of Parts tank drivetrain in season; of	Onshape CAD
■	**555 Montclair Robotics**		**34**	#987	18-24	3-stage cascade elevator with horizontal scissor 'stinger', printed pn	Onshape CAD
■	**3737 Roto Raptors**	BB Mantis	**20**	#2221	12-16-2	'BB Mantis': cord-extended arm with passive claw, through-robot handli	Drive folder / doc
■	**9062 Critical Circuits**	Bryce	**19**	#2368	15-34	'Bryce': long-range dual-flywheel cube shooter, two side roller intake	Onshape CAD
	663 Power Sharks	MAKO	**18**	#2546	11-12	'MAKO': 6 NEO 6WD; electric chain-driven elevator, pneumatic arm exten	Onshape CAD

■ = study pick: downloadable CAD or a full technical binder, and one specific design lesson. EPA is Local EPA (approx), computed from The Blue Alliance match data. Records are win-loss-tie totals from official events.

Top-30 robots by Local EPA

■ **Study pick** **5940 BREAD** Pain d' Epi — USA-CA — **Local EPA 74 (#3)** — 40-9

Pain d' Epi: tilted Dyneema elevator; cube/cone floor intake whose spring-loaded cone roller sets the handoff orientation

SCORING (CONE / CUBE) End effector receives cones from the floor intake in a set orientation. Grip design unconfirmed (see CAD).	**INTAKE** Floor intake for cubes and cones. The beta robot had a cube-only floor intake; the cone roller was added later. The cone layout rotates the 3 in polycarbonate roller about the intake pivot and adds a 2 in flex wheel shaft that spins opposite the other rollers. Surgical tubing pulls the 2 in roller outward; when a cone is in far enough the rollers lock and the 2 in roller pivots about the 3 in roller to a hard stop. The team states another motor would have put the robot over the weight limit.
ARM / ELEVATOR Tilted elevator, adapted from a 2022 off-season project elevator. That project tested a belted elevator; the team chose Dyneema instead because it was easier to package, routed through the tubes, and the belt gearbox took more space.	**DRIVETRAIN** Swerve: the team lists swerve offsets among possible causes of the same playoff failure. Module type unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD)	**AUTONOMOUS** The team added cone floor pickup so auto could place 2 cones and a cube for a high link, instead of a preloaded cone and two cubes. They expected many Championship matches to tie at 193 points and be decided by auto. They state supercharging later reduced the value of this. At Central Valley, a practice-match auto failed; AdvantageKit log replay showed a wrong heading target, traced to how the code flipped it.
VISION AprilTag vision: the team lists it among possible causes when the robot lost control in a division playoff match. Camera count and software unconfirmed (see code).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Keep all layout sketches in one Onshape document and derive them into each subsystem document.
- Try passive geometry before adding a motor: locked rollers make the cone roller pivot and orient the cone for the handoff.
- Off-season projects can feed the season: a 2022 project elevator went onto the 2023 robot, mounted tilted, which the team says helped them build quickly.
- Base mechanism scope on match math: the team added cone pickup expecting tied Championship matches to be decided by auto points.
- Log everything and replay it: at Curie playoffs the logs showed no gyro data, which let the electrical team fix the robot before the next match.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers video

Profile	6329 The Bucks' Wrath Shock USA-ME	Local EPA 74 (#2) 55-14

'Shock': two-position tipping elevator; dual-motor intake with a wrist; SDS MK4i drivetrain; 24 x 24 in frame

SCORING (CONE / CUBE) Dual-motor intake with a wrist, per the reveal post. The wrist is driven by Falcon motors; its position comes from their internal encoders after the robot starts each match in the same position. Roller layout unconfirmed.	**INTAKE** Dual-motor intake with a wrist (see scoring). The Behind the Bumpers summary mentions in-fill intake wheels.
ARM / ELEVATOR Two-position tipping elevator, per the reveal post. Stage count and drive are not stated by the team; the Behind the Bumpers summary describes a two-stage cascading elevator. Elevator extension is driven by Falcon motors and measured with their internal encoders.	**DRIVETRAIN** SDS MK4i modules, per the reveal post. Frame is 24 in x 24 in, set by the team as a design rule.
ENDGAME Unconfirmed (not stated in the public release)	**AUTONOMOUS** Routines are not described in the public posts. Autonomous Award at the Rhode Island event.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Reveal thread (CD)

LESSONS FOR STUDENTS

- A self-imposed frame size can drive a design: the team set 24 in x 24 in as a priority rule, smaller than its usual robots.
- The elevator tips between two positions, and a wrist on the intake adds a second degree of freedom at the end effector.
- Motor internal encoders can be enough for position control: the team starts the elevator and wrist in the same place each match and reads the Falcon internal encoders, a scheme it reused on its 2024 robot.

CD thread: Reveal thread (CD) · Reveal thread (CD) · Behind the Bumpers (CD) · Team post on the 2023 elevator and wrist control (2024 thread, CD)

■ Study pick

2056 OP Robotics UPPERCUT

Canada

Local EPA 73 (#4)

61-8

'UPPERCUT': MK4i swerve; arm with gas-shock shoulder hold; driver-set cone/cube intake modes; upward Limelight

SCORING (CONE / CUBE) End effector type unconfirmed (see the binder). The robot had no colour sensor; the driver set cone or cube mode (see intake). LEDs flash white when a game piece is acquired. This tells the driver the robot holds it when the intake is out of view.	**INTAKE** Intake with side rollers and a top roller. The driver chose cone or cube mode, because the side roller speeds and the top roller position differed by game piece. In cone mode a beam break sensor detected the cone and the intake clamped onto it automatically. In cube mode the rollers stopped when the cube hit a bumper switch at the back of the intake. The OP names 'single substation floor pickups' as a case where the driver cannot see the intake or the game piece.
ARM / ELEVATOR Arm on a shoulder pivot; layout details unconfirmed (see the binder). Gas shocks went over centre to hold the shoulder pivot in the starting configuration. The top pivot point attaches to 1/8 in box beam. On the practice bot a crack formed there and the tube wall deformed. The team added 1/4 in aluminum plates with four #8 bolts. The OP names Loctite 609 as the retaining compound used to remove backlash.	**DRIVETRAIN** Swerve. The team released STL covers for its swerve modules, and the file name names MK4i. The covers kept foreign material out; the team removed the top cover to inspect each module before and after each event. To grease a module, the team put grease on the end of a zip tie, fed it into the gearbox with the middle cover on, and turned the wheels. A Pigeon 2 gyro showed no noticeable drift over a match.
ENDGAME Unconfirmed (not stated in the public release).	**AUTONOMOUS** Routines not described in the thread. The OP reports that Phoenix Pro time sync improved the accuracy and repeatability of the odometry. Autonomous Award at the Archimedes Division and at ONT District Newmarket.
VISION One Limelight in the drivetrain, angled up to track cone nodes and AprilTags. No game piece detection in auto or teleop; the team states it did not need it.	**PUBLIC CAD / DESIGN** Technical binder (PDF)

LESSONS FOR STUDENTS

- Use the practice robot to find structural failures: the top pivot mount on 1/8 in box beam cracked on the practice bot, and the team added 1/4 in aluminum plates and four #8 bolts as a precaution.
- Time-synchronized sensor data: the OP credits Phoenix Pro time sync for better odometry accuracy and repeatability.
- Intake states without a colour sensor: the driver sets cone or cube mode, a beam break triggers the cone clamp, a bumper switch stops the cube rollers, and LEDs flash white on acquisition.
- Gas shocks set over centre held the shoulder pivot in the starting configuration.
- Module covers as a reliability part: the STL files are public, the team inspected each module under its cover before and after each event, and greased the gearbox through the middle cover with a zip tie.

Tech binder: Technical binder (PDF) · Technical binder release thread (CD) · Technical binder (PDF) · Swerve module covers (STL zip) · Behind the Bumpers thread (CD) · Behind the Bumpers video

■ Study pick

2910 Jack in the Bot Phantom

USA-WA

Local EPA 73 (#6)

69-16

'Phantom': 26x28 robot under 17 in stowed, 2-stage cascade-rigged telescoping arm, floor and shelf pickup; SDS Mk4i L3 swerve

SCORING (CONE / CUBE) The robot scores on all three levels (reveal post). A team reply describes the cone sitting in the intake and the robot ejecting it to score, so the intake also acts as the end effector. Roller and motor detail: Unconfirmed (see CAD).	**INTAKE** Picks up cones and cubes from the ground and the shelf, including fallen-over cones (reveal post). A Time of Flight sensor on the inner wall of the intake measures where the cone sits. Mechanism detail: Unconfirmed (see CAD).
ARM / ELEVATOR 2-stage cascade-rigged telescoping arm (reveal post) on a shoulder pivot (team reply). Green cord on pulleys on the first stage drives the second stage from the first stage's motion (OP reply). A team reply names an A-frame structure with a dead-axle main pivot as part of the low-backlash design. Joints use only the integrated Falcon encoders, with REV ION gears for a tight fit on hex shafts (team reply).	**DRIVETRAIN** SDS Mk4i L3 swerve on a 26x28 frame perimeter (reveal post). A team reply states three gearings were run in 2023: standard L3, L3 with a 16T pinion, and L3 with an 18T pinion. The robot weighs about 90 lbs without ballast and is under 17 in tall when stowed.
ENDGAME Auto balancing on the Charge Station (reveal post). Dock and engage method: Unconfirmed (see code).	**AUTONOMOUS** 3 game piece autonomous and auto balancing, listed as separate features (reveal post). Whether one routine does both: Unconfirmed. Paths come from PathPlanner. The team did not use the PathPlanner marker system; custom code runs commands in parallel or in sequence with path following (code thread).
VISION Two angled cameras, used constantly in teleop to relocalize the robot on the field (team reply). Camera model and coprocessor: Unconfirmed (see code).	**PUBLIC CAD / DESIGN** 'Phantom' CAD (Onshape)

LESSONS FOR STUDENTS

- Sequence arm joints to limit tipping: the code does not extend the arm until the shoulder is within an allowed distance of its setpoint.
- A telescoping second stage can be driven without its own motor: cord fixed to the base stage runs over pulleys on stage one and pulls stage two.
- Low backlash without absolute encoders: the team used only Falcon integrated encoders, REV ION gears with a tight hex fit, and a dead-axle main pivot.
- Driver-assisted scoring: a Time of Flight sensor finds the cone center, the robot PIDs to the nearest column, and it ejects only when X, Y, yaw and pitch pass strict thresholds. The team reports the driver was often faster.
- Path-following feedforward can be tuned by hand: the team turns off path PID, tunes velocity feedforward on a slow straight path, then acceleration feedforward, and repeated this for each of three drive gearings.

Onshape CAD: 'Phantom' CAD (Onshape) · 2023 CAD and tech binder release (CD) · Robot reveal (CD) · 2023 code release (CD) · Code (GitHub) · Tech binder (PDF) · Behind the Bumpers video

■ Study pick

3005 RoboChargers

USA-TX

Local EPA 73 (#5)

58-12

Tower-mounted chain-driven arm with a pivoting, turnbuckle-tensioned gearbox; wrist per Behind the Bumpers; see Onshape CAD

SCORING (CONE / CUBE) The Behind the Bumpers video description lists a fully articulating wrist. Gripper and roller details are not stated in the release thread; unconfirmed (see CAD).	**INTAKE** The Behind the Bumpers video description lists an 'on the fly' intake. Mechanism details are not stated in the release thread; unconfirmed (see CAD).
ARM / ELEVATOR Chain-driven arm on a tower. The arm gearbox plates react off an aluminum cross tube at the base of the tower, and 2 turnbuckles tension the chain. The mount is a slip fit on the tube with collars holding it axially, so the whole gearbox pivots around the tube. The team says this also gives some side-to-side play to adjust chain tracking. A reply in the thread mentions an Onyx-printed e-chain support on a telescoping part of the arm.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Charge Station approach not described; unconfirmed (see CAD).	**AUTONOMOUS** Routines and piece count not stated; unconfirmed.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Tension a chain by pivoting the whole gearbox around a round tube, set with 2 turnbuckles. The team took this from heavy equipment practice: it makes it easier to keep the shafts parallel, and rotational motion typically needs fewer and simpler parts than a linear adjuster.
- One cross tube does two jobs: it is structure for the tower and the pivot for the gearbox. The team says they wanted the cross member for structure anyway.
- The team expected to adjust chain tension often because of the force from the fast arm, so the adjustment is part of the mount.
- The release is the working model, not a cleaned-up copy. The team presents it as an example of a well-organized but practical FRC CAD model.

Onshape CAD: Onshape CAD · CAD release (CD) · Onshape CAD · Behind the Bumpers (CD) · Behind the Bumpers video

Profile	4414 HighTide Tsunami USA-CA	Local EPA 71 (#8) 60-10-1

'Tsunami': slanted elevator with a v4b and intake; before Chezy Champs changed to a 1323-style arm and wrist

SCORING (CONE / CUBE) At the season events the robot used a v4b with an intake. Changed before Chezy Champs: the v4b and intake were replaced with a 1323-style arm and wrist, built in the team's own style with the same roller locations. The new arm and wrist are made from 7075 and light weighted to reduce mass.	**INTAKE** Intake plates are 7075-T6 aluminum; the team says the plates took big hits without damage, though a front roller snapped. Rollers are 1.25 in polycarbonate tube with 1 in silicone tubing stretched over it, with Markforged-printed hubs glued in, on 3/8 in ThunderHex. The team added star wheels after seeing a pair on 1323's intake. The intake pivot runs on a single Falcon at about 30:1 in Motion Magic Voltage mode.
ARM / ELEVATOR Slanted elevator with WCP inline elevator blocks and a tensioned belt last stage. The team names worn constant force springs as the only major elevator issue; it was tensioned once on the competition robot. Before Chezy Champs the team added ballast low on the back of the elevator to prevent tipping.	**DRIVETRAIN** Drivetrain type is not stated by the team. The Behind the Bumpers summary mentions custom swerve modifications; module type unconfirmed (not stated in the public release).
ENDGAME Unconfirmed (not stated in the public release).	**AUTONOMOUS** During the season the robot scored two mid cones and one high cone in auto. After the Chezy Champs changes it scored two high cones without tipping. The Behind the Bumpers summary mentions on-the-fly dynamic trajectories. Autonomous Award at Ventura County.
VISION Changed before Chezy Champs: a tower was added to hold a Limelight for more reliable retro tape tracking, plus a rear-facing Limelight for cone tracking in auto. The team says a rear camera was hard to fit on the low side of the slanted elevator before the change. Season vision setup unconfirmed.	**PUBLIC CAD / DESIGN** Reveal and design Q&A thread (CD)

LESSONS FOR STUDENTS

- Mechanical stiffness reduces tuning work: the team says PID effort approaches zero when a mechanism is light and has low backlash.
- Hex gear backlash can be removed with .002 in shim stock tape wrapped around 1-3 faces of the bore; the team posted photos of the process.
- Position subsystems avoid belts except for a tensioned last stage; the team prefers well-shimmed gears and chain.
- Mass placement matters for auto: ballast low on the back of the elevator let the robot score two high cones without tipping.
- The team builds without a lathe, cutting shafts on a cold saw and spacing them with COTS spacers and standoffs.

CD thread: Reveal and design Q&A thread (CD) · Reveal and design Q&A thread (CD) · Reveal video · Behind the Bumpers (CD) · Behind the Bumpers video

■ Study pick

111 WildStang

USA-IL

Local EPA 70 (#16)

38-5-1

Lift, belt-driven arm and wrist with a cone-wedging gripper; double substation loading; REV MAXSwerve; two cross-eyed Limelights

SCORING (CONE / CUBE) A gripper on a wrist scores cones and cubes. The team says the gripper wedges on the funnel shape of a cone. After a cone trap in a Seven Rivers match, the team added a plastic closeout plate to the front of the robot. The team reports 11-12 teleop cycles after its Championship upgrades, scoring on the middle set of links.	**INTAKE** Loads at the double substation from the human player. Before the Championship, the team reviewed its season match video and found that substation pickup time was its biggest slowdown. Changed for the Championship: cones were fed on their side, an idea the team credits to team 324. The scoring auto-align code was reused to set the pickup pose at the substation. Floor pickup: Unconfirmed (see CAD).
ARM / ELEVATOR A lift with constant force springs, a belt-driven arm and a wrist with an encoder. The team says the arm is backlash sensitive and uses MAX Planetary gearboxes. A wrist encoder broke from an impact at Seven Rivers; the team then supported the encoder axle on both ends. Rigging capstans are 3D prints sleeved in aluminum so rigging tension does not crush them.	**DRIVETRAIN** REV MAXSwerve swerve. The team bought five modules in fall 2022 because SDS MK4i supply was uncertain; it ran SDS MK3 in 2021 and 2022. A team member says the 2023 drivetrain used NEOs. The team zeros modules in the REV fixture with only the bevel fork installed, and machined clearance pockets into the fixture.
ENDGAME The team mentions balancing at the end of a Seven Rivers playoff match. Dock and engage method: Unconfirmed (see code).	**AUTONOMOUS** Added before the Championship: a 3-piece auto on the bump side. The team says this kept it compatible with 7-game-piece alliance autos. A team member says the team uses its own path follower, which takes velocity, acceleration and direction from the path and handles rotation separately. TBA lists the Autonomous Award at the Seven Rivers Regional.
VISION Two Limelight cameras in a 'cross-eyed' layout. The team says the second camera widened the field of view so a nearby AprilTag was always in sight, instead of relying on wheel odometry. Changed for the Championship: both cameras were turned up to see the loading station tag, about 35 deg cross-eyed and 10 deg above the horizon. Camera model unconfirmed.	**PUBLIC CAD / DESIGN** 111 2023 CAD (Onshape)

LESSONS FOR STUDENTS

- Find where cycle time goes: the team reviewed its season match video, found substation pickup was its biggest slowdown, and redesigned that step.
- Use the shape of the game piece: cones fed on their side let the gripper wedge on the cone, which the team says gives a larger target.
- Protect sensors from impact loads: after a wrist encoder broke at Seven Rivers, the team supported the encoder axle on both ends.
- Two cross-eyed cameras widen the field of view, so a nearby AprilTag stays in sight at scoring spots where one camera would lose it.
- Split the controls by role: the driver chose when each robot action happened, and the operator chose what that action did. A team member says other students picked up driving easily at the fall driver tryouts.

Onshape CAD: 111 2023 CAD (Onshape) · 111 2023 CAD (Onshape) · 2023 build blog (CD) · 2023 code (GitHub) · 2023 code release post (CD build blog) · Behind the Bumpers video

■ Study pick

2075 Enigma Robotics Cypher

USA-MI

Local EPA 69 (#18)

55-15

'Cypher': arm with shoulder and elbow joints; ground cube intake; Limelight localization in auto; drivetrain unconfirmed

SCORING (CONE / CUBE) Unconfirmed (see CAD). The Behind the Bumpers summary mentions a pass-through and a beater bar; the team posts do not describe the end effector.	**INTAKE** Ground cube intake, improved after the Muskegon event per the team update. Geometry and motors unconfirmed (see CAD).
ARM / ELEVATOR Arm with shoulder and elbow joints. After Muskegon the team reduced backlash in both joints with clocked steel gears and Loctite 680 between gear bore and hex shaft. The top joint uses custom torque-transmission parts; the team considered MAXSpline and did not use it.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The team update states they optimized the drivetrain using Phoenix Pro.
ENDGAME Unconfirmed (see CAD)	**AUTONOMOUS** New autonomous routines after Muskegon, shown in a team video, with Limelight localization. Autonomous Awards at Muskegon and at the Michigan State Championship division.
VISION Limelight localization for autonomous, added after the Muskegon event.	**PUBLIC CAD / DESIGN** Onshape CAD (publication)

LESSONS FOR STUDENTS

- Backlash fix for gears on hex shafts: clocked steel gears plus Loctite 680 to fill the gap between gear bore and shaft.
- Prototype speed can drive part choice: the team made custom top-joint parts instead of MAXSpline to use parts on hand.
- Iterate between events: the team update, posted after its second district event, lists six changes made since its first event, from drivetrain tuning to carry handles.

Onshape CAD: Onshape CAD (publication) · Reveal and CAD/code release thread (CD) · Onshape CAD (publication) · Code (GitHub)

Profile	1577 Steampunk Jesko Israel	Local EPA 68 (#19) 51-12

'Jesko': arm with telescoping extension; ground intake for tipped cones added after week 1; drivetrain unconfirmed

SCORING (CONE / CUBE) Unconfirmed (not stated in the public release)	**INTAKE** Ground intake added after ISR District Event #1 to pick up tipped cones without losing the existing intake function. A team mentor states it also makes multi-cone autos easier. Two outer bearings pass through the plates with shoulders facing out; the inner bearings sit inside with shoulders facing the 5 mm polycarbonate plate, which floats about 1.35 mm between them. The shaft is tapped and closed with end screws. The team names 95, 6328 and 2230 as inspirations.
ARM / ELEVATOR Arm with a telescoping extension. Both use Motion Magic with gravity feedforward. A team planner moves them together and adds at most 2 safe waypoints before the target to avoid the robot and the grid. A spring on the extension helps when the arm is vertical and pulls a cone, and speeds retraction and pickup; in horizontal mode it works against extension.	**DRIVETRAIN** Drivetrain unconfirmed (not stated in the public release).
ENDGAME Unconfirmed (not stated in the public release)	**AUTONOMOUS** A team mentor states the ground intake makes it easier to score multiple cones in auto. Routines are not described. Autonomous Award at ISR District Event #1.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Reveal thread (CD)

LESSONS FOR STUDENTS

- Floating bearing mount: the plate floats about 1.35 mm between bearing shoulders, so polycarbonate flexes without bolt holes for the bearings.
- Coordinate two joints with a simple planner: the team skipped a full motion profile and added at most 2 safe waypoints.
- Add a mechanism mid-season without losing existing function: the ground intake came after week 1, and the team states it did not hurt the existing intake capabilities.
- A spring can help in one pose and resist in another: the extension spring helps vertical cone pulls but works against horizontal extension.

CD thread: Reveal thread (CD) · Reveal thread (CD) · Behind the Bumpers thread with team replies (CD) · Season recap thread (CD)

■ Study pick

6036 Peninsula Robotics Dum-E

USA-CA

Local EPA 68 (#21)

49-18-1

'Dum-E': arm and ground intake with chain drives; cameras moved to the rear mid-season; drivetrain unconfirmed

SCORING (CONE / CUBE) Unconfirmed (see CAD)	**INTAKE** Ground intake. Its chain tensioner is cantilevered off the outer gearbox plate; the team states the chain load was low enough for this to work. They state that in the future they would probably consider a larger center-to-center distance to add an inline tensioner. A rectangular cutout in the intake gearbox held the camera mounts used at the first 2 events.
ARM / ELEVATOR Arm with a chain run and idler sprockets. The team states the cantilevered tensioning likely caused excess wear on the standoffs the idler sprockets spin on, though not enough to need replacement. Their off-season redesign used tensioners inspired by team 33.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Unconfirmed (see CAD)	**AUTONOMOUS** Routines are not described in the public posts. Autonomous Award at Idaho and Autonomous Action Award at CalGames.
VISION Cameras (type unconfirmed); moved from the intake gearbox to the back of the robot after the first 2 events for better field of view	**PUBLIC CAD / DESIGN** GrabCAD CAD

LESSONS FOR STUDENTS

- A cantilevered chain tensioner can work at low load (ground intake) but likely wore the idler standoffs at higher load (arm), per the team.
- Leave center-to-center distance for an inline tensioner: the team states it would probably consider this change.
- Plan camera placement for field of view: the team moved its cameras from the intake gearbox to the rear after 2 events.

GrabCAD: GrabCAD CAD · CAD release thread (CD) · STEP CAD (Google Drive) · GrabCAD CAD · Behind the Bumpers (CD)

Profile

125 NUTRONs

USA-MA

Local EPA 66 (#23)

62-25-3

Multi-stage telescoping arm (52 in of extension per a team reply) with wrist and claw; cube intake; drivetrain unconfirmed

SCORING (CONE / CUBE) Wrist and claw, per the Behind the Bumpers summary. A team member states two SPARK MAX controllers sit in the end effector. The wrist and intake NEOs run on 30A protected circuits. Gripper geometry is unconfirmed (see the video).	**INTAKE** The Behind the Bumpers summary names a cube intake. Its pickup location (floor or substation) and roller layout are unconfirmed (not stated in the public release).
ARM / ELEVATOR Multi-stage telescoping arm, per the summary. A team member states the arm is 23 in long fully collapsed, with 52 in of linear extension beyond that. End-effector wires run up the arm through a 4-wire power slip ring and a 4-wire CAN slip ring. A helix drum, driven off the arm extension gearbox, spools the cable; surgical tubing on about 8 in of cable keeps tension.	**DRIVETRAIN** Drivetrain unconfirmed (not stated in the public release). On 2023-02-23 a team member gave the weight as 78 lb, with a 26 lb powder-coated steel belly pan still to be added as ballast.
ENDGAME Charge Station dock and engage approach unconfirmed (not stated in the public release).	**AUTONOMOUS** Auto routines unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Behind the Bumpers thread (CD)

LESSONS FOR STUDENTS

- Power and CAN reach the end effector of a telescoping arm through two separate 4-wire slip rings, per a team member.
- A cable drum with a helix, driven off the arm extension gearbox, pays out and takes up the wires as the arm moves. The team states the drive ratio took some tweaking.
- A tension element absorbs tolerance and direction changes in the cable path. The team used surgical tubing on about 8 in of cable.
- Ballast is part of the design: the robot weighed 78 lb before a 26 lb steel belly pan was added, per a team member.

CD thread: Behind the Bumpers thread (CD) · Behind the Bumpers thread (CD) · Behind the Bumpers video

■ Study pick

254 The Cheesy Poofs BREAKDOWN

USA-CA

Local EPA 66 (#28)

36-5

'BREAKDOWN': rear vertical elevator, tilted 3-stage 'Laterator', one-side roller claw for cones and cubes; MK4i swerve

SCORING (CONE / CUBE)
An all-in-one roller claw intakes and scores both cones and cubes from the same side of the robot. The team says it is lightweight and centers cones from the feeder station. The claw wheels are on dead axles: 3D printed pulleys with wheels pressed, glued or bolted on, with embedded 3/8 in ID bearings that spin on 3/8 in round standoffs held between the plates.

INTAKE
Floor intake that deploys linearly on 3D printed Markforged bushing blocks lined with thin teflon tape, in place of a 4-bar. The team credits 125's early robot reveal for the idea. The floor intake roller uses the same dead-axle construction as their 2022 robot. A Hall sensor on the retracted hard stop zeros the intake. A 10DP metal gear on a 3/8 in hex jackshaft drives a rack cut into the inner of two 1/4 in polycarbonate side plates. The rack skipped a few times. The team thinks that in a hard impact the plate bends up and the teeth skip over the gear; a pit reset took about 30 seconds. The team recalls that by Championship it reduced the plate clamp bolts to 2, so the plates could flex like leaf springs.

ARM / ELEVATOR
A vertical elevator at the back gives an 'up and in' pose at the feeder station; the team says most other teams had to approach the station slowly because of an extended arm or angled elevator. The elevator carries a 3-stage 'Laterator' with cascade rigging, tilted slightly upward to reach the high cone nodes. The elevator runs on 3/4 in OD, 1/4 in ID radial ball bearings. The Laterator uses 3/8 in OD, 1/4 in ID PEEK bushings rolling on dowel pins to save weight over steel ball bearings. The Laterator tubes are carbon fiber. The team believes many Laterator breaks came from holes drilled through the tubes. Screws thread into billet end plugs or into 1/8 in aluminum 8-32 nut plates slid inside the tube. The 10-32 screws that join the elevator uprights to the A-frame came loose often; the team names them as a source of lost lateral and torsional stiffness and does not recommend the joint.

DRIVETRAIN
Swerve on MK4i modules with Falcon motors; the team supported the Falcon shaft, as in 2022. Custom steel lower L-plates with a chamfer put weight down low and clear the Charge Station. CNC-milled wedges on the wheel yokes smooth the impact of driving over the cable protector. The battery sits in the middle of the swerve base. The robot weighed 115 lb without battery or bumpers and about 138 lb with both.

ENDGAME
A forks carriage for Charge Station balancing; a Hall sensor stops the fork motor when the forks are fully climbed. The team says that by Championship and Chezy Champs it found less use for the forks, because most drivers could triple balance fast. The forks stayed on to add weight at the back and reduce forward tipping.

AUTONOMOUS
A 3-piece auto on the bump (cable protector) side; the wheel wedges were added to make it more reliable. The elevator starts midway up on a small kickstand for a faster first placement. The team says same-side intake and scoring made auto harder, because the robot must spin while driving.

VISION
Unconfirmed (not stated in the public release). For game piece sensing, the team says it used Banner DQ12AB6LV retroreflective sensors in 2023, as in other years.

PUBLIC CAD / DESIGN
Technical binder (PDF)

LESSONS FOR STUDENTS

- Architecture as a chain of decisions: the team traces the rear vertical elevator, the tilted 3-stage Laterator and the all-in-one claw back to one choice, driving straight into the feeder station for a repeatable cycle.
- The team names the cost of its layout: intaking and scoring on the same side made swerve autonomous harder, because the robot had to spin while driving.
- Weight and failure in a telescoping stage: the 3-stage Laterator rolls on 3/8 in OD PEEK bushings on pressed dowel pins in place of steel ball bearings. The team believes holes drilled through its carbon fiber tubes caused many of its breaks.
- Printed pulley practice: after a printed 18T 5mm HTD pulley sheared in 2022, the team used more aluminum pulleys and printed the rest in solid fill with about four 1/16 in shear-pin holes.
- A linear floor intake on 3D printed bushing blocks with teflon tape, which the team says may replace a 4-bar as its default. Its polycarbonate rack skipped teeth in hard hits; the team would cut teeth into both plates for full gear face width and extend the bearing block over the mesh.

Tech binder: Technical binder (PDF) · Technical binder and code release (CD) · Technical binder (PDF) · Code (GitHub) · Behind the Bumpers video · Behind the Bumpers thread (CD) · CAD progression post (CD)

■ Study pick

2767 Stryke Force Snake Eyes

USA-MI

Local EPA 66 (#29)

52-14

'Snake Eyes': carbon fiber elevator pivoted by printed sector gears, arm with bevel-gear claw; printed custom swerve

SCORING (CONE / CUBE) Claw with bevel gears in the end effector; how the gears drive the claw is unconfirmed (see CAD). A team member lists larger bevel gears in the claw as a change made between St. Joseph and Worlds. The team's later off-season post says they experimented with printed bevel gears on the 2023 end effector.	**INTAKE** The OP mentions cone intake from the floor with the smaller arm. A separate floor intake is not described; unconfirmed (see CAD).
ARM / ELEVATOR Single-stage carbon fiber elevator. Printed Onyx sector gears pivot the elevator, and a printed Onyx gearbox on the elevator carriage moves the arm. The code lists a superstructure (shoulder) twist, left-right. The team added the elevator stage to reduce tilt of the first axis, shorten the second axis by a few inches and reduce the moment arm off the base. Before Worlds they fitted a larger, taller curved rack, which they call the 'Hedgehog', and made the elbow gearboxes more durable.	**DRIVETRAIN** Custom swerve, mostly printed in Onyx, on a chromoly steel frame. A team member gives the initial weight as about 100 lb, with a 20 lb plate added under the belly pan for stability.
ENDGAME Charge Station approach not described in the release; unconfirmed (see code).	**AUTONOMOUS** Routines and piece count not stated; unconfirmed (see code). The team says the elevator-top camera was added for AprilTag viewing in auto.
VISION PhotonVision for AprilTag detection, and automatic game piece placement using AprilTags, per the code release. A camera was added on top of the elevator before Worlds for a better view of AprilTags in auto. A light on the robot turns on during camera auto-alignment; a team member says this makes video review easier.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Printed composite parts in load paths: the arm gearbox, the elevator sector gears and most of the swerve modules are Onyx. Study which loads the team gave to printed parts.
- The team put an elevator stage under the arm to reduce first-axis tilt, shorten the second axis by a few inches and reduce the moment arm off the base.
- Some carbon fiber joints are glued after final assembly: holes in the tubes lead to grooves that the team fills with hot glue.
- Onshape workflow for a student CAD team: master sketches set the layout, each subassembly gets its own folder and document, one person owns the top assembly, and gear profiles come from FeatureScripts.
- The release thread lists the changes from St. Joseph to Worlds: a taller curved rack, more durable elbow gearboxes, an elevator camera and larger claw bevel gears.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Robot reveal 'Snake Eyes' (CD) · Behind the Bumpers video

Profile	3467 Windham Windup USA-NH	Local EPA 66 (#25) 55-17-4

Double-jointed arm (per interview summary) with pneumatic claw and wrist; pneumatic cube shooter; SDS MK4i L2 swerve

SCORING (CONE / CUBE)
A claw on a wrist; the Behind the Bumpers summary mentions several iterations of both. The claw is closed by a pneumatic cylinder for cones and is open for cubes; the cylinder is spring-loaded normally open. 3D-printed pulleys with aluminum inserts ran on the claw all season without a failure. The competition robot also carried a pneumatically deployed cube shooter that collects and scores cubes; the practice robot did not have one.

INTAKE
30A 4 in Thrifty compliant wheels. The team reports they gripped cones and cubes well, and chose them for price and the softest durometer. The team lists $7.50 per Thrifty wheel against $9 for WCP and $10 for AndyMark. The team kept them in service all season with Rubber Rejuvenator and never replaced them. The claw picks up from the Double Substation, and the cube shooter also collects cubes. The Behind the Bumpers summary (2023-02-28) mentions a new intake planned for Week 1; its design is unconfirmed.

ARM / ELEVATOR
Double-jointed arm, per the Behind the Bumpers interview summary. A Behind the Glass video of a Week 0 playoff match shows the drive team working through arm issues. Changed at the World Championship: pocketed steel claw gears were replaced with pocketed aluminum, saving about 0.44 lb and making the arm more stable in fast moves. The team also replaced MAXPlanetary 1/2 in hex output stages that gained slop, and re-shimmed a 24T hex sprocket on its shaft. Joint motors and ratios unconfirmed.

DRIVETRAIN
Swerve on SDS MK4i modules with L2 gearing. The team states drivetrain speed limited them only at the World Championship. Swerve code was derived from Team 364's BaseFalconSwerve. Changed at the New England District Championship: the rear steel drive rail went back to aluminum and the two side drive rails changed to steel. The team made this change after the robot tipped over the back drive rail at Greater Boston, and to improve the center of gravity.

ENDGAME
The Behind the Bumpers summary mentions auto balance. Mechanism and routine detail unconfirmed.

AUTONOMOUS
Path following with the PathPlanner application and library. At the District Championship the robot ran a 2.5 piece auto, and the team worked toward 3 pieces. Before Worlds the team worked to restructure its autos around the 2023 PathPlanner Event Markers and Path Groups. Its existing autos did not use them and lost about a second between each path and action. The Worlds 3 piece attempt used the cube shooter for the second and third pieces, and the team reports it got close. The team debuted a 3-piece autonomous mode at the BattleCry off-season event. Autonomous Award at NE District North Shore.

VISION
Unconfirmed (not stated in the public posts). In November 2022 the team wrote it would likely use retro-reflective tape targeting in 2023 instead of AprilTags; later posts do not say what ran.

PUBLIC CAD / DESIGN
2023 build blog (CD)

LESSONS FOR STUDENTS

- Ballast placement sets tip behavior: after tipping over the back rail, the team moved steel from the rear drive rail to the two side rails.
- Fix weak COTS parts at the source: the team replaced stock two-piece welded Falcon 500 shafts with one-piece 316 stainless 8mm keyed shafts after welds broke.
- Mass at the end of an arm matters: aluminum claw gears in place of steel saved about 0.44 lb and made fast arm moves more stable.
- Pick the failure state of a pneumatic: at Worlds a fitting failed and the cube shooter stayed down and cracked, but the normally open claw still scored cubes from the Double Substation.
- Sequenced autos can lose time between commands: without PathPlanner Event Markers and Path Groups, the team's autos lost about a second between each path and action.

CD thread: 2023 build blog (CD) · 2023 build blog (CD) · 2024 build blog with 2023 recap (CD) · Behind the Bumpers thread (CD) · Behind the Bumpers video · Behind the Glass video (Week 0)

Other profiled robots

Each has public CAD, a binder, code or a detailed release thread. Sorted by Local EPA. ■ marks the study picks.

Profile	7157 µBotics Tai Lung USA-CA	Local EPA 65 (#33) 33-11-1

'Tai Lung': 65 deg elevator with an arm (one Falcon, 80:1); fork buddy climb; drivetrain unconfirmed

SCORING (CONE / CUBE) The intake and its carriage weigh 16 lbs together, per the team. Behind the Bumpers highlights the intake and wrist. Roller layout and wrist drive are unconfirmed (see CAD).	**INTAKE** Unconfirmed (not stated in the public release). Floor and substation pickup are not described (see CAD).
ARM / ELEVATOR The team states the elevator is at 65 degrees to the horizontal, with its top about 70 in from the ground. The arm is about 19 in from pivot to the farthest point. The arm runs off a Falcon through a MAXPlanetary 20:1 gearbox and a 64:16 sprocket reduction, 80:1 overall.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD).
ENDGAME Fork buddy climb. On 2023-03-28 the team posted a video of a successful iteration and planned to use it at Aerospace Valley. Charge Station dock and engage method is unconfirmed.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- The team gives a full arm reduction on one Falcon: MAXPlanetary 20:1, then 64:16 sprockets, 80:1 overall. Use it with the stated 16 lb intake and carriage as a sizing reference.
- The fork buddy climb went through iterations in season. The team posted a test video of the working version before Aerospace Valley.
- The team answered design questions in the reveal thread with numbers that the CAD alone does not state, such as the 16 lb intake and carriage weight.

Onshape CAD: Onshape CAD · Onshape CAD · Robot reveal thread (CD) · Fork buddy climb test video (Google Drive) · Behind the Bumpers thread (CD)

■ Study pick

5460 Strike Zone VIPER

USA-MI

Local EPA 64 (#44)

64-28

'VIPER': SDS MK4i swerve; Falcon intake that hands off cones; 2-Falcon carbon fiber arm; claw with Onyx-printed gears

SCORING (CONE / CUBE) Claw powered by a Falcon at 8.4:1 with Thrifty Bot wheels; 2.7 lbs at the reveal. By the State Championship the team cut a half pound from the end effector and relocated its motor to reduce rotational mass. All claw gears and pulleys are printed in Onyx filament to keep the arm's swinging inertia low.	**INTAKE** Deployed by a Falcon at 53.3:1 and driven by a Falcon at 3:1. The intake orients cones and cubes for a handoff to the claw. As of 2023-04-04 the team did not hand off cubes, because testing put the arm in vulnerable positions; it took cubes from the shelf to score high or mid.
ARM / ELEVATOR Arm powered by 2 Falcons at 140:1, with 3D printed joints and aluminum supports. The distal arm changed to 120:1 by the State Championship. The arm uses carbon fiber tubes from Clearwater Composites. Part 5460-FRC23-201-D07 began as two Onyx prints joined with 1/8 in dowel pins and epoxy. The prints split along layer lines, so the team changed the part to laser sintered Nylon 12.	**DRIVETRAIN** 26 in x 26 in on SDS MK4i L2 swerve modules, changed to L3 by the State Championship. After Championship the team added two Falcons for a 10-motor swerve: two 3-motor modules (L1 gears, 24t pinion) and two 2-motor modules (L3, 18t pinion), about 24 fps free speed. A relocating plate for the 18t pinion was machined from 3/8 aluminum. The CAD shows the robot as it was at the World Championship, before this change.
ENDGAME Unconfirmed (not stated in the public release).	**AUTONOMOUS** Routines unconfirmed (not stated in the public release). Autonomous Award at FIM District Kettering University Event #1.
VISION Two Limelights first updated odometry from AprilTags, placed so one would always see at least two tags when scoring. This allowed zone-based gyro locks at the grid and the player station. The team dropped it after its week 3 Ontario event for lack of tuning time. It then used Limelight tracking of the vision tape on the node poles.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Mass at the end of an arm: Onyx-printed claw gears and pulleys, a half pound cut from the end effector, and a relocated claw motor all reduce the arm's swinging inertia.
- Material change after a failure: a two-piece Onyx print split along its layer lines and was replaced with a solid laser sintered Nylon 12 part.
- Scope cuts with stated reasons: the team stopped cube handoffs to protect the arm and dropped AprilTag alignment when it could not tune it in time.
- An offseason drivetrain test: 10-motor swerve with mixed gearing, a follower motor per module, and a driver-station switch to disable the extra motors; two brownouts in 26 matches.
- The CAD is a SOLIDWORKS export to Onshape, and the team warns that some parts may be missing or appear incorrect.

Onshape CAD: Onshape CAD · Onshape CAD · CAD release thread (CD) · Reveal thread with specs (CD) · Reveal video · Behind the Bumpers video

■ Study pick

6672 Fusion Corps TARS

USA-TX

Local EPA 62 (#64)

55-19

'TARS': swerve; pivoting arm with forearm and wrist gearboxes; live-roller cone and cube floor intake; single driver

SCORING (CONE / CUBE) The Behind the Bumpers summary describes an intake that pinches cones and cubes, moved by a multidirectional 'bicep' arm. A separate scoring end effector is not described; see CAD.	**INTAKE** Floor intake with live top and bottom rollers covered in tubing. The team states the floor pickup geometry makes the bottom roller touch a cone before the top roller, so the bottom roller must be live. The design took inspiration from the 3847 and 8177 Chief Delphi threads.
ARM / ELEVATOR Pivoting arm with forearm and wrist gearboxes. The lead programmer recalls that the wrist went from 100:1 (5:1, 5:1, 4:1) to three 3:1 stages plus a 2:1 sprocket, 54:1 total, to speed up the wrist and cut MAXPlanetary lash. The arm base gearing was also reduced for speed; exact ratios are in the code Constants file. Bent polycarbonate guards absorb hits from the arm if it goes astray.	**DRIVETRAIN** Swerve, run by custom swerve code built on the team's 2022 code, with wheel optimization and a closed-loop ramp limit to prevent wheel slip. The Behind the Bumpers summary mentions a 40lb ballast. Module type unconfirmed (see CAD).
ENDGAME Charge Station approach unconfirmed (not stated in the public release).	**AUTONOMOUS** PathPlanner paths, with steps that let the team modify paths separately on each side of the field. Piece counts unconfirmed. Autonomous Award at the FIT District Fort Worth Event.
VISION A Limelight was mounted but never used. A Logitech webcam was meant to give the driver a view of the intake; the team says the driver never used it. No game piece detection.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Keep design history in the CAD: subsystem versions named .0 reached production and versions named .5 did not.
- Document gear ratio changes: the wrist went from 100:1 to 54:1 to gain speed and reduce MAXPlanetary lash.
- Arm control without inverse kinematics: setpoint moves follow an algorithm that keeps the arm on a safe path through the robot.
- Tie roller choice to geometry: the bottom roller touches the cone first, so both intake rollers are live.
- The release credits its sources (3847 and 8177 intake threads, 3015 PathPlanner) and includes a BOM.

Onshape CAD: Onshape CAD · 2023 CAD and code release (CD) · Onshape CAD · Code (GitHub) · Bill of materials (Google Sheets) · Behind the Bumpers video

■ Study pick

3357 COMETS Kushida
USA-MI

Local EPA 61 (#79)
63-20

'Kushida': 21x26 in chain-in-tube tank drive; 3-stage elevator with a geared arm; floor-first intake

SCORING (CONE / CUBE) The CAD renders show a separate end effector on the arm. Grip method, rollers and motors are unconfirmed (see CAD).	**INTAKE** Floor intake; the Behind the Bumpers summary calls the design floor-intake-first. A team member answered in the reveal thread that it picks up cones upright, tip-towards, and within about 70 degrees each way from tip-towards. Base-towards cones take two steps: flip to vertical, then intake. The team member said upright and tip-towards are the fastest.
ARM / ELEVATOR An elevator with a base stage, a 2nd moving stage and a 3rd output stage carries an arm. Wiring runs in energy chain 'A' from the base to the 2nd stage, then energy chain 'B' to the 3rd stage. The arm had no counterbalance and was geared at about 200:1 for the first events. Thin-wall tube bent a little near the elevator slots from the arm's inertial loads; the team reports no loss of function.	**DRIVETRAIN** Tank drive with chain-in-tube assemblies that the team says can be hot-swapped in 3 minutes. The team chose it over swerve for module supply, cost, limited swerve experience, and a day-9 practice drivetrain goal. Black nitrile tread, held to the wheel with safety wire stitches, lasted multiple events. Drive motor count unconfirmed (see CAD).
ENDGAME The team says the drivetrain was built to cross the Charge Station many times per match. Dock and engage method unconfirmed.	**AUTONOMOUS** TBA lists the Autonomous Award at FIM District Calvin University and FIM District West Michigan. Routines and piece counts are unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** GrabCAD CAD (STEP)

LESSONS FOR STUDENTS

- Drivetrain choice from stated constraints: the team lists module supply, cost, swerve experience and a day-9 practice drivetrain goal as the reasons for tank drive.
- Design for repair: the chain-in-tube drive assemblies hot-swap in 3 minutes, per the team.
- Motor standardization: the team kept NEO 550 motors with UltraPlanetary gearboxes on many actuations for weight, needed many revisions to make them reliable, and recommends all-NEO or all-Falcon for other teams.
- Arm loads reach the structure: a first light-weighted steel arm drive gear deformed, and the thin-wall elevator bent a little near its slots.
- A simple part numbering scheme: Assembly-Subassembly-PartNum1-PartNum2, then the part name and the author's initials.

GrabCAD: GrabCAD CAD (STEP) · 'Kushida' CAD (GrabCAD, STEP) · Reveal and CAD release thread (CD) · Behind the Bumpers thread (CD) · Behind the Bumpers video · Reveal video

■ Study pick

341 Miss Daisy Miss Daisy XXI

USA-PA

Local EPA 59 (#105)

42-19-1

'Miss Daisy XXI': belt-driven telescoping arm; roller-claw intake on a 120-deg locking wrist; SDS MK4i; 23x30 in frame

SCORING (CONE / CUBE) Roller-claw style intake on a wrist with 120 deg of movement and a locking mechanism. The lock first used a 'comb' moved by servos. The servos failed, so the team changed to a small pneumatic piston. The team scores cubes and cones at all 3 levels.	**INTAKE** The team collects cones from the single substation, the double substation, and from the floor when the cone lies on its side. Cubes are also collected; the cube pickup points are not stated (see CAD).
ARM / ELEVATOR Belt-driven telescoping arm, per the reveal. Arm motors, ratios and extension length are not stated in the public posts (see CAD).	**DRIVETRAIN** SDS MK4i modules with Falcons for drive and steering. Frame is 23 in wide x 30 in long. Robot weight is 95 lbs, per the reveal.
ENDGAME The team chose a 23 in x 30 in frame to better fit on the Charge Station with 2 full-size partners. The balance method is not described (see code).	**AUTONOMOUS** Auto routines are not described in the public posts. Unconfirmed (see code).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- The endgame can set the frame size: the team built a 23 in x 30 in frame to better fit on the Charge Station with 2 full-size partners.
- A wrist lock moved by servos destroyed the servos. The team replaced them with a small pneumatic piston. Check actuator loads before you choose a servo.
- One roller-claw intake takes cones from the single substation, the double substation and the floor on their side. Compare that claim with the geometry in the CAD.
- CAD, code and a design binder come from the same release, so you can trace one robot from design notes to model to software.

Onshape CAD: Onshape CAD · CAD, code and binder release (CD) · Onshape CAD · Code (GitHub) · Robot reveal thread (CD)

Profile

695 Bison Robotics Goldmodule

USA-OH

Local EPA 59 (#109)

52-11

'Goldmodule': cascading elevator with an Everybot intake (per Behind the Bumpers); swerve code

SCORING (CONE / CUBE) Unconfirmed (not stated in the public release; see CAD).	**INTAKE** Behind the Bumpers describes an incorporated Everybot intake. Rollers, motors and pickup points are unconfirmed (see CAD).
ARM / ELEVATOR Cascading elevator, per the Behind the Bumpers description. Stage count, motors and rigging are unconfirmed (see CAD).	**DRIVETRAIN** Swerve, per the team's code repository description ('2023 Swerve Code for Team 695'). Module type is unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release; see code).	**AUTONOMOUS** Unconfirmed (not stated in the public release; see code).
VISION Unconfirmed for the 2023 robot. The release post lists an off-season swerve and vision code rewrite as a separate project.	**PUBLIC CAD / DESIGN** CAD (Google Drive, STEP/IGES)

LESSONS FOR STUDENTS

- Behind the Bumpers describes a cascading elevator with an incorporated Everybot intake. Use the CAD to study how a published intake design was fitted to the team's own elevator.
- The team developed its code in public on GitHub, with conventions carried over from the Rapid React season, per the release post.
- The CAD is released as STEP/IGES files, a neutral format that any CAD package can import.

Drive folder / doc: CAD (Google Drive, STEP/IGES) · CAD and code release (CD) · CAD (Google Drive, STEP/IGES) · Code (GitHub) · Behind the Bumpers thread (CD)

■ Study pick

1706 Ratchet Rockers Whiplash

USA-MO

Local EPA 59 (#111)

38-11-1

'Whiplash': cone and cube shooter on swerve; frame corners cut to the module size; Limelight driver assist

SCORING (CONE / CUBE) Cone and cube shooter. The release post states the team set out to shoot the game pieces instead of placing them, for a higher possible skill ceiling, and reached a working robot after a lot of trial and error. Motors, ratios and wheels are unconfirmed (see CAD).	**INTAKE** The Behind the Bumpers summary describes dual intakes. Intake type and layout are unconfirmed (see CAD).
ARM / ELEVATOR Unconfirmed (not stated in the public release); see CAD.	**DRIVETRAIN** Swerve; the team calls its modules 'mk2'. The frame corners are cut to the module size. The team says this saved about 6 in of corner-to-corner diameter so the robot can rotate off defense more easily.
ENDGAME Charge Station approach unconfirmed (not stated in the public release).	**AUTONOMOUS** Routines unconfirmed (not stated in the public release); see the 2023 code. The team won the Autonomous Award at the Central Missouri Regional.
VISION Limelight. A team member states the frame shape did not work as well as hoped for lining up on the grid, so the team used the Limelight to give the driver assistance on approach.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A shooter can replace a placing arm in a placing game: the team chose to shoot cones and cubes for a higher possible skill ceiling.
- Cutting the frame corners shrinks the swept circle: the team cut its frame to its 'mk2' swerve module size and reports about 6 in less corner-to-corner diameter.
- Check design intent against the team's own statement: the octagonal frame was not a grid-alignment feature; the team used Limelight driver assistance for alignment.

Onshape CAD: Onshape CAD · 2023 code and CAD release (CD) · Onshape CAD · Code (GitHub) · Robot reveal 'Whiplash' (CD) · Behind the Bumpers video

■ Study pick

4795 Eastbots HYDRA

USA-NC

Local EPA 59 (#106)

50-15-1

'HYDRA': mid-level scorer; roller claw on a wrist; roller bar cube intake added in season; no high cones

SCORING (CONE / CUBE) A roller claw on a wrist. The claw side plates are 1/8 in polycarbonate. During the season the team removed the aluminum scab plates and shortened the claw ('stumpy', replacing 'lanky'); the team credits the shorter claw for better cone pickup at the double substation. Dead-axle pulleys run on 3/16 in bore radial bearings (R3-2RS). The wrist has a pivot and a piston mount. The team states HYDRA could not score high cones and focused on the middle level.	**INTAKE** The Behind the Bumpers summary after the UNC Asheville event names a 'flail intake'. The team describes a roller bar intake, with mecanum rollers and a pressure-set deploy, as an addition during the season; it hands cubes to the claw. The team says it added 2-3 game pieces to a typical teleop and sped up feeder cycles. After the state championship the team added top hardstops for cubes and increased the mecanum roller density.
ARM / ELEVATOR Arm or elevator geometry is unconfirmed (see CAD). The Behind the Bumpers summary describes the robot as compact. At the state championship the team swapped a spare wrist and claw unit that had the wrong spacers; it bound in a match until the holes were drilled to clearance size.	**DRIVETRAIN** Drivetrain unconfirmed (see CAD). The team's recap describes strafing motions during pickups; the drive type is not named.
ENDGAME 'Landing gear' for the Charge Station, described in the Behind the Bumpers video. Details unconfirmed (see CAD).	**AUTONOMOUS** Routines and piece counts are unconfirmed. The team reports that roller bar cube pickups often failed in auto but worked in teleop. The team added small strafing and wiggling motions to every auto pickup, and the problem mostly stopped. The team says auto consistency was its weak point at the state championship.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Limit the task list to what the team can do well: the team states HYDRA did not score high cones and focused on the middle level, and later named 'build simple but effective' as its motto.
- Iterate the end effector on a measured problem: shortening the roller claw improved double substation cone pickup, per the team.
- Debug the difference between auto and teleop: cube pop-outs happened only in auto, and the team's hypothesis was the absence of driver motion during pickup.
- Write failure reports: the team documented a mis-spaced spare wrist and sheared electronics shelf standoffs, with the fix for each.
- CAD, code and a technical binder were released together, so mechanism, software and documentation can be read side by side.

Onshape CAD: Onshape CAD · Onshape CAD · Technical binder (Google Doc) · Code (GitHub) · 2023 build thread (CD) · Reveal 'HYDRA' (CD) · Behind the Bumpers video · 2024 Open Alliance thread with the 2023 release post (CD)

■ Study pick

8177 Vector Velocity

USA-TX

Local EPA 58 (#112)

36-24-2

'Velocity': MK4i L3 swerve, 55 deg tilted elevator, NEO wrist, triple-roller intake for tipped/standing cones and cubes

SCORING (CONE / CUBE) The triple-roller intake on the wrist is also the scoring end effector. The rollers turn in opposite directions for cones and cubes; the code sets the outtake direction from the last intake position. Intake hubs on the wrist axle are printed in Carbon Fiber PLA at nearly 100% infill, with printed TPU disks to spread the clamp load on the polycarbonate intake panel.	**INTAKE** Triple-roller design that takes tipped cones, standing cones and cubes (the team's programming update). In February testing the intake still took cones with the bottom roller about 1 in off the ground. LEDs flash yellow or purple to ask the human player for a cone or a cube.
ARM / ELEVATOR Tilted elevator at 55 deg from the horizontal with double chain rigging, a TTB constant force spring and WCP bearing blocks. The wrist rotates the intake on 1 NEO at 32:1, with a REV Through Bore encoder for absolute position. As of March 2023 the elevator had no absolute encoder due to supply issues; it returns to the same start position.	**DRIVETRAIN** Swerve on MK4i L3 modules. The chassis is 26 in x 26 in with a steel belly pan; the team states the drive chassis weighs 60 lbs by itself for a low CG. One-piece bumpers.
ENDGAME Balance is part of the 1+balance auto path (March 2023). Teleop Charge Station dock/engage approach unconfirmed (not stated in the public release).	**AUTONOMOUS** PathPlanner trajectory following, the team's first season with it, using AprilTag and swerve encoder pose. As of 2023-03-20 the team ran 1+1 and 1+balance paths and planned a 1+1+1+balance path.
VISION A Limelight reads AprilTags, and the pose estimate combines this with the swerve drive encoders. A held driver button drives the robot to the grid and leaves the driver only left-right control to line up. An OpenCV cone-centering test was in early development in March 2023.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A constant force spring on the elevator countered gravity enough that the team held the elevator with a P-only loop and no feedforward.
- The wrist needed SysId feedforward (S, V, G, A). The team states gains calculated off the robot were inaccurate, and it switched gains when a cone added weight.
- Setpoints sequence the elevator and wrist moves. The team rewrote the move order after the intake caught on the level 2 cone pole.
- Driver load reduction: one score button serves cones and cubes, because the code infers the game piece from the last intake position.
- Low CG by mass placement: the 26 in x 26 in chassis with a steel belly pan weighs 60 lbs by itself.

Onshape CAD: Onshape CAD · 2023 build thread (CD) · Onshape CAD · Code (GitHub organization) · OpenCV cone-centering test (GitHub) · Behind the Bumpers video · Robot video (March 2023)

■ Study pick

2337 EngiNERDs Vision

USA-MI

Local EPA 57 (#145)

45-17-1

'Vision': shoulder, elbow and wrist arm with rope-driven elbow and cable-driven wrist; NEO 550 intake; SDS MK4i L3 swerve

SCORING (CONE / CUBE) Wrist end effector. Wrist: Falcon with MaxPlanetary (4:4:4), brake-cable driven from the belly of the robot. Intake rollers: NEO 550 with UltraPlanetary (5:4). Grip geometry unconfirmed (see CAD).	**INTAKE** NEO 550 with UltraPlanetary (5:4), per the spec list. Floor or substation pickup is not stated; see CAD.
ARM / ELEVATOR Arm with shoulder, elbow and wrist. Shoulder: Falcon, MaxPlanetary (5:5:4), RT-25 belts (72:32). Elbow: same motor and reductions, rope driven from the shoulder pivot. Wrist as described under scoring.	**DRIVETRAIN** Swerve on SDS MK4i modules, L3 ratio, per the spec list.
ENDGAME Unconfirmed (see CAD)	**AUTONOMOUS** Unconfirmed (not described in the team posts).
VISION Unconfirmed (not described in the team posts).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Remote joint drives: the elbow is rope driven from the shoulder pivot and the wrist is brake-cable driven from the belly.
- Shoulder and elbow share one reduction stack: Falcon, MaxPlanetary 5:5:4, RT-25 belts 72:32.
- Belly pan hardware: asked about snagging on the Charge Station, the team says it typically puts button head cap screws or rivets on the bottom of the belly pan, or uses zip ties.

Onshape CAD: Onshape CAD · Reveal, spec list and CAD release thread (CD) · Onshape CAD · Behind the Bumpers (CD)

Profile

177 Bobcat Robotics The Bobcat

USA-CT

Local EPA 54 (#191)

48-19

'The Bobcat': double-jointed arm on an angled elevator; swerve (the team's first swerve robot)

SCORING (CONE / CUBE) End effector type unconfirmed (see CAD).	**INTAKE** Unconfirmed (see CAD). The Behind the Bumpers video summary says the intake went through several iterations.
ARM / ELEVATOR The team describes the 2023 robot as 'essentially a double-jointed arm mounted to an angled elevator'. After the 2023 season, one student derived the kinematic equations for this layout with geometry and trig as a preseason project. The goal was to set scoring and intake positions more directly.	**DRIVETRAIN** Swerve. The team states that 2023 was its first year with a swerve drivetrain. The 2023 code used Team 364's BaseTalonFXSwerve code. Module type unconfirmed (see CAD).
ENDGAME Unconfirmed (not stated in the public release).	**AUTONOMOUS** Routines unconfirmed (see code). The team reports problems with its 2023 autos that it could not debug. This led it to add AdvantageKit logging for 2024.
VISION Limelights with AprilTags, chosen as the simplest option for the time available. The team later reported limited field of view and slightly inaccurate pose estimation due to the mounting.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- A team student derived kinematic equations for a double-jointed arm on an angled elevator with basic geometry and trig. Check them against the joint layout in the CAD.
- A first swerve season on a published base: the 2023 code used Team 364's BaseTalonFXSwerve instead of a new code base.
- Camera mounting affects pose estimates. The team reports limited field of view and slight pose error from its 2023 Limelight mounting.
- Two code repositories (Bot 1 and Bot 2) are public for the same season.

Onshape CAD: Onshape CAD · Onshape CAD · Code, Bot 1 (GitHub) · Code, Bot 2 (GitHub) · 2024 build blog with 2023 links (CD) · Reveal thread (CD) · Behind the Bumpers thread (CD)

■ Study pick

3847 Spectrum X-Ray 2023

USA-TX

Local EPA 54 (#182)

44-20

'X-Ray 2023': cube-shooting swerve robot (per Behind the Bumpers) with an elevator, slide and arm/intake; see CAD

SCORING (CONE / CUBE) Cube shooting robot, per the Behind the Bumpers video title and description. The January sketch used a cone pinch intake that doubles as a ball (cube) intake, with a flywheel or roller to launch cubes in auto. Final cone and cube end effector unconfirmed (see CAD).	**INTAKE** The team unbolted the slide, arm and intake to ship the robot, and planned to carry a spare arm/intake from its practice robot on the plane. The kickoff plan was to load cones from the Human Player shelf and pick cubes from the floor. Final intake geometry unconfirmed (see CAD).
ARM / ELEVATOR Elevator with a slide and arm. In January 2025 replies in the build blog, the team named external cantilevered pulleys as the main problem. By the end of 2023 these ran on ~2.75 in shoulder bolts through tube end caps, which still bent and sheared. The team also named the lower elevator mounting as a weak point.	**DRIVETRAIN** Swerve. In 2022 the team mounted MK4i modules below the frame rails so a whole module could drop out. It never swapped a whole module at an event. For 2023 it used a standard module mount, because modules below the frame rails would have raised the belly pan and electronics. The 2023 module type is unconfirmed (see CAD).
ENDGAME Charge Station approach unconfirmed (not stated in the public release). At an off-season event the post does not name, a bumper came off and the robot drove to the Charge Station before it could be disabled. Another user described the play; the team credits its drive coach for the call.	**AUTONOMOUS** Behind the Bumpers describes PathPlanner routines. The team posted a Chezy Champs 'Bump' side auton mode on 2023-09-29. Piece counts are unconfirmed.
VISION The robot carries a camera rail that unbolts for shipping. Camera hardware and software are unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Team CAD site (Onshape)

LESSONS FOR STUDENTS

- Kickoff trade study: the team chose shelf cones over floor cones. It states the shelf sits a bit higher than the mid cone node and that 15 of 27 nodes accept cubes.
- Failure note: cantilevered pulleys need high belt tension. By the end of 2023, ~2.75 in shoulder bolts through tube end caps still bent and sheared.
- Design for shipping: the slide, arm, intake and camera rail unbolt and lie on the chassis without undoing wiring, except the RSL. The robot fit a 48x40x38 freight box.
- Requirement review: the 2022 drop-out module mount was never used at an event, so the team dropped that requirement in 2023 for a lower CG.
- Software structure: the team's fall software plan makes every function, including auton and vision, its own subsystem, with template classes for rollers, arms and elevators. The team wrote that not all of it was implemented yet.

CAD release: Team CAD site (Onshape) · Build Blog 2023 (CD) · Team CAD site (Onshape) · Code site (GitHub) · Design review of X-Ray (video) · Behind the Bumpers video

■ Study pick

955 Ctrl C

USA-OR

Local EPA 53 (#212)

44-28

Roller claw on an extension that cannot reach the floor; ground cube intake added by DCMP; first-year swerve; 95.5 lb

SCORING (CONE / CUBE) Roller claw on the extension mechanism. The team says it held game pieces securely. Motors and wheel details are unconfirmed (see CAD).	**INTAKE** No ground intake at the district events. Ground cube intake was added by DCMP and the Championship. Earlier the team pursued a 'weedwacker' ground intake based on 2158's prototypes; the team says it was never going to work and that it spent too long on it. Pickup method at the district events is unconfirmed (see CAD).
ARM / ELEVATOR Extension mechanism that carries the roller claw. The team describes it as fast and simple, with few failures under impact loads. It was designed so it could not reach the ground, because the team assumed a ground intake would be too heavy to extend. Extension type is unconfirmed (see CAD).	**DRIVETRAIN** Swerve, the team's first year with it; the team says it would not have done as well without it. Module type is unconfirmed (see CAD). Weight was 95.5 lbs at SunDome against 124 lb for the 2022 robot, mostly from 1/16 in box tubing for most of the structure.
ENDGAME Unconfirmed (not stated in the public release).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Check early assumptions: the team assumed a ground intake was too heavy to extend, so it designed an extension that could not reach the floor and a handoff to a roller claw. The team says this made later improvements hard.
- Stop work on a prototype that is not converging: the team says it spent too much time forcing a 'weedwacker' ground intake that never fully worked.
- Weight control by material choice: the robot weighed 95.5 lb against 124 lb in 2022. The team says this is mostly from 1/16 in box tubing for most of the structure.
- Plan wiring and sensors with the other subteams: the team names missing wiring plans and design-programming miscommunication about sensors as a negative.

Onshape CAD: Onshape CAD · 2023 CAD release (CD) · Onshape CAD

■ Study pick

2881 Lady Cans SCoRpion

USA-TX

Local EPA 53 (#222)

34-26

Two-lead-screw pivoting and extending arm with a suction-cup end effector; swerve

SCORING (CONE / CUBE)
Two suction cups near the end of the arm hold the cone or cube. The team chose them over a claw with compliance wheels after prototyping, because they carry less weight at the end of the arm. Early tests wrinkled the cube under vacuum; the team reports this was solved. The vacuum line under the arm is pneumatic tubing inside heat shrink, coiled around a PVC pipe with a heat gun so it acts like a spring.

INTAKE
The first plan used hex-shaft rollers covered in silicone tubing. In build week 2 the team changed the plan to an intake like the Everybot intake. A four-bar linkage keeps the roller section parallel to the ground. It picks up both game pieces and drops them inside the robot for the arm, and no part of it rests on the floor. The team says the intake is the main difference between the build week 4 CAD and the final robot. Final intake design unconfirmed.

ARM / ELEVATOR
Arm driven by two lead screws. One lead screw moves the arm up and the other extends it outward. The team started the season with a 0.5 in per turn lead screw, the same screw and 3:1 reduction as its 2022 robot. Changed after testing and the first event to a 1 in per turn lead screw to cut cycle time; the team notes it back-drives more, so it may need a brake or more software. Motors unconfirmed (see CAD).

DRIVETRAIN
Swerve. The team states 2023 was its first year with swerve in official competition. The four modules came from its 2022 off-season swerve bot. The robot was light, so the team planned a solid steel drivetrain to add weight and lower the center of gravity. Module type unconfirmed (see CAD).

ENDGAME
Buddy balance mechanism, still being machined in build week 5. Mechanism detail unconfirmed (see CAD).

AUTONOMOUS
Trajectory following with PathPlanner, first tested on the off-season swerve bot. In build week 2 the robot could not check that it held a game piece before scoring, which affected preload choice. Routines unconfirmed.

VISION
AprilTags, first year of use. In build week 2 the cameras recognized tags and output an estimated position. The plan was to combine this with odometry. Camera hardware unconfirmed.

PUBLIC CAD / DESIGN
GrabCAD CAD (to build week 4)

LESSONS FOR STUDENTS

- A suction-cup end effector is a lighter option than a claw at the end of a long arm; the team chose it for that reason after prototyping both.
- Two lead screws give the arm two motions, pivot and extension. The team doubled the lead screw pitch after its first event to cut cycle time, and reports the faster screw back-drives more.
- In build week 2 the intake plan changed to a four-bar that stays parallel to the ground and hands pieces inside the robot, so nothing rides on the floor. The team says the final intake differs from the week 4 CAD.
- A coiled vacuum line can be made in the shop: the team put pneumatic tubing inside heat shrink and set the coil on a PVC pipe with a heat gun.
- The GrabCAD file stops at build week 4. The team states the intake is the main difference from the final robot, so compare the CAD with the build thread.

GrabCAD: GrabCAD CAD (to build week 4) · 2023 build thread (CD) · GrabCAD CAD (to build week 4) · Behind the Bumpers video · 2024 build thread with 2023 recap (CD)

■ Study pick

401 Copperhead Robotics Mushussu

USA-VA

Local EPA 52 (#251)

42-21-1

Double-sided arm (250-deg rotation, 30 in extension) with a 330-deg belt-in-tube wrist; SDS MK4i L2 swerve

SCORING (CONE / CUBE) The end effector grabs a cone by its flange or tip from any orientation, from both sides. The robot scores cones and cubes out either side. Roller or claw type is not stated in the release post (see CAD and tech binder).	**INTAKE** Not described in the release post beyond the any-orientation cone grab. Floor and substation pickup are unconfirmed (see tech binder).
ARM / ELEVATOR Arm with 250 deg of rotation and 30 in of extension, per the release post. A belt-in-tube driven wrist gives 330 deg of rotation.	**DRIVETRAIN** SDS MK4i L2 swerve modules.
ENDGAME Unconfirmed (not stated in the release post; see tech binder).	**AUTONOMOUS** Unconfirmed (not stated in the release post; see tech binder).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- An arm with 250 deg of rotation and 30 in of extension, plus a 330 deg belt-in-tube wrist, lets the robot score out either side. Compare this range of motion with single-sided arms.
- The end effector grabs cones by the flange or the tip from any orientation. Study the CAD to see how one gripper handles both grip points.
- Track robot enable time with an hour meter on a toggleable PDH slot.
- Toggle switches set brake or coast mode for all motors and turn all LEDs on or off.
- The same thread has Onshape CAD for a battery cart (9 slots, 3x Dual Pro RS3 chargers, plasma cut 12 gauge steel) and a driver station with split-arcade joysticks.

Onshape CAD: Onshape CAD · CAD and tech binder release (CD) · Onshape CAD

■ Study pick

7461 Sushi Squad

USA-WA

Local EPA 52 (#256)

36-25

Chain-driven pivot arm with pneumatic extension and a claw; intake with indexer handoff; Falcon 500 swerve

SCORING (CONE / CUBE) A claw on the end of the arm. The team shared Onshape documents for the alpha claw and the competition arm and claw in the build thread on 2023-03-13. Before the first event the team found claw problems with pulleys and advises against 18T PLA pulleys. In March the team posted cube shooting tests and asked if shooting cubes is a viable strategy. Claw actuation unconfirmed (see CAD).	**INTAKE** January plan: an over-the-bumper cube ground intake with indexing wheels that hands off to the arm, plus single substation cone pickup. The team planned to ignore tipped cones. The alpha intake used pneumatics. Before the first event the team fixed indexing and handoff problems: wheels did not spin because of pulley wear. Competition intake geometry unconfirmed (see CAD).
ARM / ELEVATOR Pivot arm with a pneumatic extension. The team states the arm uses a large dead axle, #35 chain kept in tension and a large sprocket. The arm never flips, so backlash stays on one side; the team reports +/- 1-2 deg. Changed before the first event: a Spartan tensioner replaced a turnbuckle that lacked tension, and a steel hex shaft replaced aluminum to reduce bending. The absolute encoder pulley went to 1:1, and a fixed-bearing idler replaced a double-bearing floating idler that caused 2-3 deg of error. After the first event the team added quick exhaust valves to speed the extension, with retraction valves next. Arm motor unconfirmed (see CAD).	**DRIVETRAIN** Swerve, with Falcon 500 motors on the modules per the team's electrical write-up. The January plan named 2x1 tube swerve for competition; the alpha robot reused a 2x2 off-season chassis. The 3D-printed swerve corners crushed after two events. The team reprinted them with 4 walls and 40 percent infill and carries spares. Module make unconfirmed (see CAD).
ENDGAME Charge Station auto-balance. In January tests, wheels set in an X shape held on polycarbonate with almost no slip. The first bang-bang auto-balance sometimes overshot, so in March the team began testing a PID-based version. The team advises that only one robot auto-levels at a time.	**AUTONOMOUS** On 2023-03-09 the robot ran a two-piece auto on the side without the cable bump ('non-berm side', in the team's words). By the first event the team had berm-side and 'normal side' autos with balance. The team posted a 3-piece auto on 2023-04-04. Path tool unconfirmed.
VISION Vision measurements and wheel odometry feed a WPILib pose estimator for odometry-based auto-align to the grid. The team weights vision heavily because odometry drifted during defense. Auto-align for the Double Substation was planned in March. Camera hardware and software unconfirmed.	**PUBLIC CAD / DESIGN** Onshape CAD (comp robot)

LESSONS FOR STUDENTS

- Arm accuracy comes from the drive path: the team reports +/- 1-2 deg with tensioned #35 chain. Before its first event it added a Spartan tensioner, a steel hex shaft, a 1:1 encoder pulley and a fixed-bearing idler.
- Feedforward can follow the arm state: the team schedules two kG values, one with the arm piston extended and one with it retracted.
- The REV Through Bore absolute encoder did not integrate with the SPARK MAX for this team; they read it through DIO.
- Keep the CAD main assembly current: late electrical belly pans and informal design reviews led to bad battery wire bends and poor PDH access.
- Drive the robot hard before events: long runs found a PDH fire, Falcon issues and a ripped CAN termination three weeks before competition.

Onshape CAD: Onshape CAD (comp robot) · Onshape CAD (comp robot) · 2023 build thread (CD) · Arm design guide linked in the thread (CD) · Behind the Bumpers video

Profile	2877 LigerBots USA-MA	Local EPA 51 (#282) 25-24

Swerve; pivoting 2-stage telescoping arm with cascade rigging and constant-force springs; end effector unconfirmed

SCORING (CONE / CUBE) Unconfirmed (see CAD). In January 2023 the team prototyped a pneumatic linear pinch claw with rug-gripper material and a roller claw that pneumatics open and close. The team states the pinch claw needs no motors or wires at the top of the arm, and the roller claw allows touch-and-control. The posts do not state which claw went on the competition robot.	**INTAKE** Unconfirmed (see CAD). The team chose a rotating arm so the robot could pick up game pieces from the ground. In January 2023 the team planned a U-shaped bumper so the arm could clear the bumper as it swings.
ARM / ELEVATOR Rotating 2-stage elevator on a pivot, chosen during CAD weekend in January 2023 after the team could not make a static arm work. The largest stage is 2.5 in box tube. Constant force springs and cascade rigging move the stages, which the team states puts less load on the motors. The pivot motors sit low on the chassis with an inline chain tensioner. In November 2023 the 9mm HTD belt that extends and retracts the arm sheared in a mock match.	**DRIVETRAIN** Swerve, finalized by the build leaders on 2023-01-11. The January plan gives a 25 x 30 in chassis. The belly pan is perforated polycarbonate in two zones and solid polycarbonate in the center, because the team could not find more perforated sheet. Module type: Unconfirmed (see CAD).
ENDGAME The team chose the 25 x 30 in chassis so the robot would not need to beach on the ramp and three robots could fit on the Charge Station. Programmers had an auto leveling system working on the test chassis by 2023-01-30.	**AUTONOMOUS** Unconfirmed (not stated in the public release). Piece count and paths are not stated; auto leveling on the Charge Station worked on the test chassis in January 2023.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Settle arm geometry with skeleton sketches first: the team compared a static arm and a rotating arm and found the rotating geometry needed two stages.
- Constant force springs can share the load with cascade rigging: the team used them so the motors carry less load while the stages still move like a cascade elevator.
- Size belts for match impacts: the 9mm HTD belt that drives the telescoping arm sheared in a mock match in November 2023.
- Let the drive team practice before the modules are ready: the team fitted 3D-printed 'swerve dummies' with caster wheels to the chassis.
- Plan the frame for the endgame: the team chose a 25 x 30 in chassis so three robots could fit on the Charge Station.

Onshape CAD: Onshape CAD · Onshape CAD · 2023 Open Alliance build thread (CD) · 2024 Open Alliance thread, 2023 robot belt failure (CD) · 2023 robot flyer (PDF) · 2023 design process (PDF)

■ Study pick

4201 The Vitruvian Bots Gridlock

USA-CA

Local EPA 51 (#279)

26-14

'Gridlock': 3-stage rope-driven elevator; 180-deg virtual 4-bar with 3-roller intake; SDS MK4i L3 swerve

SCORING (CONE / CUBE) The release does not describe a separate scoring end effector. It names a 3-roller intake attached to the virtual 4-bar. How the robot places pieces on the nodes is Unconfirmed (see CAD). The code uses a state machine for scoring, intaking and stowed states.	**INTAKE** 3-roller intake on the virtual 4-bar. The team states it picks up ground cubes and cones, standing or tipped. Intake distance sensors to detect a game piece were developed but not fully used in season.
ARM / ELEVATOR 3-stage elevator with 5 ft of extension in under 1 second, per the team. It uses custom 3D printed pulleys and ropes inside 2 in x 1 in extrusions. A virtual 4-bar with fixed sprockets and chain gives 180 deg of motion.	**DRIVETRAIN** SDS MK4i L3 swerve. The drivetrain is 24 in x 24 in with a height of 4 1/4 in.
ENDGAME Not described in the release post. Unconfirmed (see code).	**AUTONOMOUS** Trajectory following with PathPlanner and 4 auto paths. Limelight pipelines adjust the chassis toward game pieces in auto. The autos also run in WPILib field simulation.
VISION Limelight pipelines find game pieces and adjust the chassis toward them in auto. AprilTag localization was developed but not fully used in season.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Ropes and custom 3D printed pulleys inside 2 in x 1 in extrusions drive a 3-stage elevator that the team says reaches 5 ft in under 1 second.
- Study the fixed-sprocket virtual 4-bar in the CAD: it gives the intake 180 deg of motion.
- The code uses a state machine to move between scoring, intaking and stowed states.
- The team simulated the superstructure, LEDs and field with autos, using WPILib simulation libraries and PathPlanner.
- The release lists features that were not ready in season: AprilTag localization, intake distance sensors and closest-node scoring. Read them as work in progress next to the competition code.

Onshape CAD: Onshape CAD · CAD and code release (CD) · Onshape CAD · Code (GitHub) · Behind the Bumpers thread (CD)

■ Study pick

1561 Roboducks Duck N' Cover V2

USA-OK

Local EPA 50 (#327)

39-20

Duck N' Cover V2: cube-only robot; pivoting arm with a cube shooter/intake; SDS MK4i L2 swerve; about 85 lb

SCORING (CONE / CUBE) Cube shooter/intake on a pivoting arm. For the final robot, the OP believes the shooter ratio is close to 1:1. The week 2 version was built from a piece of polycarbonate and two aluminum rods, with Falcons attached directly to the shooter. It was driven manually with joysticks. It popped cubes and broke chain. The week 6 version refines it.	**INTAKE** The shooter/intake on the arm collects cubes; the final robot does not handle cones. The week 1 robot had a four-bar intake with a floating roller, all aluminum, that weighed over 20 lb; the team scrapped it. The simplified week 1 robot used a ramp to funnel cubes from the single substation.
ARM / ELEVATOR Low pivoting arm, chosen to keep the robot low for fast cycles. The OP believes the arm uses a 20:1 AndyMark gearbox. For Tulsa the team added preset arm positions. The week 1 robot had a tower with a WCP telescoping arm driven by a winch; the team dropped the tower for week 2.	**DRIVETRAIN** Swerve on SDS MK4i modules in the L2 ratio, per the OP. The team considered L3 but did not have time. The week 6 robot has a smaller drive base than the week 2 robot.
ENDGAME Charge Station balance. The team added balancing for week 2 and had a working balance auto at Tulsa (Green Country Regional).	**AUTONOMOUS** Balance auto from Tulsa onward. After Tulsa the team ran a three-piece auto on the bump side.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** STEP CAD (zip)

LESSONS FOR STUDENTS

- Compute the ceiling of a narrow strategy before you build: the team found a cube-only robot could reach a theoretical 75 points.
- Weight affects acceleration: the OP states the final robot weighed about 85 lb with battery and bumpers and credits that for its acceleration.
- Scope to the time available: the week 1 intake weighed over 20 lb, the robot over 135 lb, and the team scrapped it days before competition.
- Secure CAN wiring: the team reports a CAN fault at Tulsa and a loose CAN wire that stopped the robot at Championship.

CAD download: STEP CAD (zip) · CAD release and season overview (CD) · STEP CAD (zip) · Code (GitHub) · Behind the Bumpers (CD)

Profile	4561 TerrorBytes Fulcrum USA-NC	Local EPA 48 (#356) 45-16-2

'Fulcrum': double-jointed arm on MAX Tube; Everybot-inspired intake added at States; SDS MK4i swerve

SCORING (CONE / CUBE) Unconfirmed (not described in the season recap). At the first event the team stopped scoring high cones to limit stress on the arm joints. At Wake County they scored low for links when their data predicted a win and scored high in close matches.	**INTAKE** The team first used a horizontal intake because it could be mounted and tested sooner. Two days before the FNC State Championship they fitted an intake inspired by Everybot, 2910 and 7890, aiming for a touch-it, own-it intake. The new intake required set-points tuned in the pit.
ARM / ELEVATOR Double-jointed arm. V1 used REV MAX Tube and competed all season. V2 was planned as an arm modeled on 971's 2018 'Exotica' arm but was not fitted. The shoulder sprocket and chain wore and stripped at the first event; the team planned a second sprocket to reinforce the joint. After States they added absolute encoders to the joints, replacing limit switches used to zero at match start.	**DRIVETRAIN** SDS MK4i swerve modules, bought at the end of the 2022 season. The team chose swerve for mobility and speed on the open field. In the off-season they ran 3D-printed tread of TPU 95A at 30% gyroid infill on PLA+ hubs.
ENDGAME Unconfirmed (not stated in the season recap).	**AUTONOMOUS** Unconfirmed (not stated in the season recap).
VISION Unconfirmed for the season robot. The team planned to add a Limelight in the off-season for odometry testing.	**PUBLIC CAD / DESIGN** 'Neptune' off-season robot CAD (Onshape)

LESSONS FOR STUDENTS

- Size chain-driven arm joints with margin: the shoulder sprocket stripped at the first event and bent again later, and the team named design for manufacture as its focus for the next year.
- Make wear parts fast to swap: the team plans to make changing a bent or stripped sprocket easier on future robots.
- Absolute encoders on the arm joints replaced limit-switch zeroing at match start, to make arm positions smoother.
- A late intake swap costs tuning time: the new intake two days before States meant tuning set-points in the pit, and the team decided not to change code in the pit in future.
- Ship the simpler version first: the V1 MAX Tube arm went to the programmers early and competed all season; the V2 arm was never fitted.

Onshape CAD: 'Neptune' off-season robot CAD (Onshape) · Open Alliance thread with 2023 season recap (CD) · 'Neptune' off-season robot CAD (Onshape)

■ Study pick

131 CHAOS

USA-NH

Local EPA 42 (#560)

20-25

Extender on linear sliders with a wrist and a light roller gripper; swerve; 1/8 in steel belly plates as ballast

SCORING (CONE / CUBE) Roller gripper. The team describes it as exceptionally rigid and extraordinarily light weight. They state more aggressive pocketing could have removed more weight. A REV UltraPlanetary drives the gripper, and the team reports no issues with it there.	**INTAKE** Unconfirmed (see CAD). The release post does not describe a separate floor or substation intake.
ARM / ELEVATOR An extender on linear sliders with needle rollers, plus a wrist. The team reports two needle roller failures over the season. In the previous season their GreyT Telescope had a bearing failure about every other match. The CAD shows REV UltraPlanetary gearboxes on the extension and wrist; the team found them not durable enough there. They changed the wrist to a 550 motor on a VersaPlanetary, and the season ended with an extension gearbox failure.	**DRIVETRAIN** Swerve; the team mentions the swerve modules and supports in the frame tubes. Module type unconfirmed (see CAD). Electronics hang from a brain pan, which the team does not recommend because the shield was hard to remove. After Week 1 they addressed CG issues with 1/8 in steel belly plates that also served as the shield. #10-32 button head screws hold the plates into captive lock nuts: 8 screws on the battery box plate and 4 on the other.
ENDGAME Unconfirmed (not stated in the public release). The team states the ballast mount had to hold when the robot flew over the Charge Station and landed on a cone.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** 2023 CAD release (CD)

LESSONS FOR STUDENTS

- Check gearbox loads before copying a design: the CAD shows REV UltraPlanetary gearboxes on the extension and wrist, and the team found them not durable enough there. The wrist moved to a 550 motor on a VersaPlanetary.
- The same UltraPlanetary had no issues on the roller gripper, which the team recommends as an example of a light, rigid roller system.
- Linear sliders with needle rollers on the extender had two failures in a season. The team wants the next version replaceable without removing the side plates.
- The 'Pocket Nut Holder' is an aluminum block that holds a #10-32 lock nut, with 0.5 in hole spacing. The team posted it as an Onshape Part Studio and describes flush-mount and stand-off versions.
- Plan service access early: removing the steel belly shield needed a drill, and fitting nut holders mid-season needed router-cut access holes in the tubing.

CD thread: 2023 CAD release (CD) · 2023 CAD release (CD) · 2023 robot CAD (Onshape) · Pocket Nut Holder Part Studio (Onshape) · Team code (GitHub)

Profile	4322 Clockwork USA-CA	Local EPA 42 (#571) 16-9

Swerve on SwerveX modules with Falcon 500 motors; floor pickup limited to cubes; single-substation cycles

SCORING (CONE / CUBE) Unconfirmed (see CAD). The technical binder covers the final design of each subsystem; the public posts do not describe the end effector.	**INTAKE** Floor pickup worked only for cubes; the team states this was due to its intake design. The team cycled to the single substation and looked for ways to cut that trip time. In November 2023 the team printed two new intake braces and fitted one.
ARM / ELEVATOR Arm on a pivot shaft with a chain. In November 2023 the team replaced the arm pivot shaft. The arm chain was loose after that, and the team retensioned it. Arm geometry unconfirmed (see CAD).	**DRIVETRAIN** Swerve on Swerve X modules with Falcon 500 motors. The team states that defense against the robot was a major problem during the season. In the off-season the team built 3-motor swerve modules from the OpenSwerve design with NEO steering. In December 2023 it took one apart and started to assemble a 2-motor SDS MK4i module.
ENDGAME Unconfirmed (not stated in the public posts).	**AUTONOMOUS** Unconfirmed (not stated in the public posts). The technical binder has a section on the autos.
VISION Type unconfirmed (not stated in the public posts). The technical binder has a section on the vision system.	**PUBLIC CAD / DESIGN** 2023 robot CAD (Onshape)

LESSONS FOR STUDENTS

- Compare design options in writing: the binder lists the options for each subsystem with pros and cons, and the changes made between the two regionals.
- Plan wire routes through structure: a December 2024 team post says many wires on the 2023 robot were damaged where they were crammed through a tube.
- An intake limit drives other choices: floor pickup worked only for cubes, so the team looked at drivetrain speed to cut single-substation cycle time.
- More drive motors need current limits: the team planned current and voltage monitoring because 3-motor swerve can draw more power than the breaker handles.

Onshape CAD: 2023 robot CAD (Onshape) · 2023 CAD, code and binder thread (CD) · 2023 robot CAD (Onshape) · 2023 code (GitHub) · 2023 Technical Documentation (Google Drive)

Profile	3136 O.R.C.A. USA-VA	Local EPA 41 (#600) 25-21

Tilted 3-stage elevator with intake on the middle carriage; narrow rectangular frame on traction wheels (Jan 2023 build log plan)

SCORING (CONE / CUBE) The January plan holds game pieces in a separate intake on the elevator's middle carriage. The team chose a separate intake on the middle carriage instead of using the end of the elevator as the intake, as Cranberry Alarm's Ri3D robot did. Final end effector unconfirmed (see CAD).	**INTAKE** V1 took its actuation idea from team 3636: a plate and custom gears bolted to the arm. It had a flat top for cones, a diamond-shaped area for cubes and compliant wheels; the team judged it over-engineered. V2 used 2 in belting to grab cones and compliant wheels as 'claws' for cubes. V2 held cubes well but did not grab cones consistently from above bumper height. Final design unconfirmed (see CAD).
ARM / ELEVATOR Tilted 3-stage elevator, chosen by vote from 17 concepts; the team cites Cranberry Alarm's Ri3D design. Sketch limits: first stage no longer than 56 in to fit the frame perimeter, about 35 deg to score within the extension limit, and a pivot from about 60 deg to about 35 deg. The team needed elevator angle control so a cone does not hit the poles. The team's 2024 log calls it an 'elevator on an elevator'.	**DRIVETRAIN** Traction wheels. The team rejected mecanum because it could be pushed and lose traction on the Charge Station; omni and pneumatic wheels were also rejected. Custom flipped gearboxes save space in a narrower, rectangular frame; v1 was built, then adjusted and re-cut. Motor count unconfirmed (see CAD).
ENDGAME The team chose a narrower robot so three robots can fit more easily on the 8 ft wide Charge Station for the Activation bonus. Dock and engage method unconfirmed.	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Derive mechanism geometry from the rules first: the team sketched the elevator in Onshape and found the 56 in first-stage limit and the 35 deg scoring angle before detailed CAD.
- A structured concept choice: 17 concepts from Ri3D robots and 2018-2019 robots, two rounds of voting, and a tilted elevator picked because it was simple, the team had built an elevator before, and Ri3D had documented it.
- Intake iteration with stated failures: V1 was over-engineered, and V2 held cubes but missed cones from above bumper height.
- Robot width chosen for the endgame: a narrower frame makes room for three robots on the Charge Station.

Onshape CAD: Onshape CAD · 2023 Open Alliance build log (CD) · Onshape CAD · 2024 build log with 2023 recap (CD)

Profile	1018 Pike RoboDevils USA-IN	Local EPA 39 (#687) 11-18

AM14U5 8-wheel kit chassis with NEOs; single-joint 'Sushi arm' planned for low/mid scoring and substation pickup

SCORING (CONE / CUBE) Not final in the January posts. A quick roller claw mock-up showed that cubes are easy to intake. A student's roller claw intake design did not get the cone compression right. In late January, students were working on how to integrate an Everybot-style pinch roller. Final end effector unconfirmed (see CAD).	**INTAKE** Substation shelf pickup of cones and cubes was a 'need'. Floor pickup was on the will-not-do list, though the mock-up geometry might allow floor pickup of cubes. A student mock-up of the Everybot intake picked up cones at many angles.
ARM / ELEVATOR A 'Sushi arm': a single-joint arm optimized for low and mid scoring, with influence from 7461's single-joint dead-axle arm. A passover arm was tested with a wooden mock-up and dropped because each score took about 230 deg of arm motion. A pneumatic extension for the high cone node, as on Sushi Squad's Alpha bot, was under consideration. Final configuration unconfirmed (see CAD).	**DRIVETRAIN** AM14U5 kit chassis with the 8-wheel XL Performance wheel kit and a 24 in frame width. The performance wheels kept the HiGrip track width while cutting 2 in of frame width. 10.86:1 EVO Slim 2-motor gearboxes with NEOs, chosen for a sprint distance of about 20 ft. Swerve was on the will-not-do list.
ENDGAME Dock and engage on the Charge Station in auto and endgame was a 'need'. Mechanism unconfirmed (see CAD).	**AUTONOMOUS** Planned routine: score the preloaded piece, drive across the auto line, and dock and engage. Programmers were working on auto-balance in late January.
VISION Unconfirmed. Automated AprilTag alignment was listed as a 'want' in January.	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- Write the priority list as need, want and will-not-do, with a deadline for the needs: the team set mid and low scoring, substation pickup and auto docking for week 4-5, and put floor pickup and swerve on the will-not list.
- Use a physical mock-up when CAD capacity is short: students built a wooden arm on a kit chassis to test arm length, pivot height and front-rear bias.
- Check the cost of a passover arm in motion per cycle: the mock-up showed about 230 deg of travel per score, so the team kept the arm on one side.
- Frame width is a design variable in this game: the XL Performance wheels cut 2 in of frame width at the same track width, to leave more room on the Charge Station.

Onshape CAD: Onshape CAD · 2023 Open Alliance build thread (CD) · Onshape CAD · Code (GitHub)

Profile	3739 Oakbotics Blue Steel Canada	Local EPA 39 (#697) 20-28

'Blue Steel': MAXSwerve chassis; intake redesigned after week 1 (released render shows the week 1 intake)

SCORING (CONE / CUBE) Unconfirmed (see CAD)	**INTAKE** The release render shows the week 1 intake. At the Georgian event the team judged it a bottleneck, then designed, prototyped and built a new REV1 intake over the next week. REV1 details are unconfirmed (see the image in the release post and the CAD).
ARM / ELEVATOR Unconfirmed (see CAD)	**DRIVETRAIN** MAXSwerve chassis, driving by week 2 of build season, per the release post.
ENDGAME Unconfirmed (see CAD)	**AUTONOMOUS** 1.5 piece bump side auto, run through the DCMP playoffs per the release post. Other routines: Unconfirmed (see code).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** GrabCAD CAD

LESSONS FOR STUDENTS

- Released CAD can differ from the competition robot: this render shows the week 1 intake, which the team replaced.
- Fast iteration after week 1: the team designed, prototyped and built a new intake in one week.
- The team had a driving MAXSwerve chassis by week 2 of build season and credits its programmers for the control loops and autos.

GrabCAD: GrabCAD CAD · CAD and code release (CD) · GrabCAD CAD · Code (GitHub)

■ Study pick

3926 MPArors Snowdrift

USA-MN

Local EPA 38 (#734)

14-17

'Snowdrift': low-goal cycler on a slim modified KoP tank base with EVO Shifters and 8 in pneumatic wheels

SCORING (CONE / CUBE) Low goal only. The team planned to cycle game pieces from the substation to the hybrid nodes or to alliance partners. The cycle target was 8-10 full-field cycles a match, about 12 s each. End effector detail unconfirmed (see CAD).	**INTAKE** One intake for both cones and cubes. The team rejected separate cone and cube intakes on opposite sides because two intakes on the small frame would have made it hard to mount the electronics and the battery. An early concept used four powered compliant wheels: a wider front pair for cubes and a narrower rear pair for cones. 7 in inside edge to inside edge worked for the outer black compliant wheels. The separate cube intake prototype used four compliant wheels, two stacked.
ARM / ELEVATOR No arm or elevator. The team prototyped a single-degree-of-freedom high-goal arm (inspired by 1503 and 179's 2011 robots) and dropped it. It cited the arm's constraints and its experienced leaders being spread too thin.	**DRIVETRAIN** Modified kit-of-parts drivetrain in its slimmest configuration. The team states it would run 'some form of tank drive' because its SDS swerve modules arrived too late. The team chose AndyMark EVO Shifters to cut time crossing midfield, and 8 in pneumatic wheels so it could drive over the Charge Station. Motors 'most likely' 4 or 6 NEO at the design stage; unconfirmed (see CAD).
ENDGAME The team aimed for a small, light robot so it could join a double balance and help enable a triple balance. Balance method unconfirmed.	**AUTONOMOUS** Unconfirmed (not stated in the public release). See the GitHub code.
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** 'Snowdrift' full assembly (Onshape)

LESSONS FOR STUDENTS

- A low-goal cycler can be a planned role: the team argued that a fast feeder lets a high-goal partner skip the slow, congested half of each cycle.
- Set a number before choosing parts: 8-10 cycles of about 12 s led to EVO Shifters and 8 in pneumatic wheels.
- Drivetrain plan follows the field: without swerve (its SDS modules arrived late), the team planned tank drive on pneumatic wheels to cross the Charge Station repeatedly and leave the congested lanes free.
- Cut scope to fit resources: the team dropped the high-goal arm when prototyping both mechanisms spread its experienced leaders too thin.

Onshape CAD: 'Snowdrift' full assembly (Onshape) · 2023 Open Alliance thread (CD) · 'Snowdrift' full assembly (Onshape) · Code (GitHub)

Profile	1710 The Ravonics Revolution SlipSTream USA-KS	Local EPA 37 (#821) 27-20

'SlipSTream': double-jointed arm on 6 NEOs, pneumatic claw with self-leveling pads, 2-roller intake; Mk4 swerve

SCORING (CONE / CUBE) Pneumatic claw with self-leveling pads. The team says the robot can score on all levels.	**INTAKE** 2-roller intake inside the frame perimeter, per the March 2023 reveal. The team's November 2023 CowTown report mentions an over-the-bumper intake belt that slid off and a broken pulley. The two sources differ; confirm in the CAD.
ARM / ELEVATOR Double-jointed arm driven by 6 NEOs. The team calls the gear ratio 'absurd' but does not state it. At CowTown the arm once failed to lift a game piece out of the intake in auto, and once did not respond properly to commands when a NEO wire disconnected at a junction.	**DRIVETRAIN** Mk4 swerve on an A-frame chassis, 28 in x 28 in. The robot weighs roughly 105 pounds.
ENDGAME Charge Station approach not described; unconfirmed (see CAD).	**AUTONOMOUS** Routines and piece count not stated. The CowTown report shows the auto sequence uses the arm to lift a game piece out of the intake.
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** Competition robot CAD (Fusion 360)

LESSONS FOR STUDENTS

- A double-jointed arm on 6 NEOs. Open the Fusion 360 CAD to find the reduction the team calls 'absurd'.
- A pneumatic claw with self-leveling pads, on a robot the team says scores on all levels.
- The team logs each match problem with cause and fix: a battery that started below 10V, driver controllers switched by the drive team, a NEO wire disconnected at a junction. They use a pre-match checklist and keep pre-made spare belts for belt wear.

Fusion 360 CAD: Competition robot CAD (Fusion 360) · Robot reveal 'SlipSTream' (CD) · Competition robot CAD (Fusion 360) · Reveal video · CowTown ThrowDown report (CD build thread)

Profile	8738 Slice USA-NC	Local EPA 35 (#905) 28-23

Elevator and arm robot on a Kit of Parts tank drivetrain in season; offseason rebuild to swerve with a wrist added

SCORING (CONE / CUBE) The team cut polycarbonate for its grabber mechanism in week 7 (post of 2023-03-01). The grabber type and how it holds cones and cubes are unconfirmed (see CAD).	**INTAKE** The arm had a recorded state for picking up a cone from the double substation, per the team's offseason programming write-up. The team built several intake prototypes in weeks 2-6. A floor intake on the competition robot is unconfirmed (see CAD).
ARM / ELEVATOR The team bought a single-stage Thrifty Bot elevator kit in week 2 and mounted its elevator in week 7. It also looked at a rack and pinion telescoping arm after problems with string-based designs in 2022, and printed a test rack segment. In season an arm on the elevator moved between recorded motor-angle states (stowed, high row, double substation) with positional PID on each motor. Changed in the offseason: the arm gained a wrist for 3 degrees of freedom, with code checks that keep the intake and wrist off the elevator.	**DRIVETRAIN** Kit of Parts drivetrain, built in weeks 2-5. The team first built a 20 in x 32 in KoP frame to test a narrow drivetrain, with the end pieces left uncut so it could go wider; final dimensions are not stated. The offseason write-up says the robot used tank drive in the 2023 season. Changed in the offseason: upgraded to swerve, the team's first.
ENDGAME Programming tested automatic Charge Station balancing code in week 3 and posted a clip captioned 'Successful auto Charge Station balancing'. Early field modeling led the team to expect that large robots may not fit a side-by-side triple engagement, and it took this into the design. The dock and engage approach used at events is unconfirmed.	**AUTONOMOUS** Programming debugged Pathweaver autos in December 2022, then started over with PathPlanner in week 2. It began testing the new auto code in week 5. Routines and piece counts are unconfirmed. The team won the Autonomous Award at FNC District Johnston County.
VISION Unconfirmed on the competition robot. Programming tested AprilTags on a Limelight 2+ in week 3 and researched PhotonVision.	**PUBLIC CAD / DESIGN** 2023 robot CAD (Onshape)

LESSONS FOR STUDENTS

- Prototype, then model, then CAD: the mentors asked for physical prototypes before design decisions, and the team modeled the robot in Blender in week 4 before CAD.
- State-based arm control: in season each arm position was a recorded set of motor angles, reached with positional PID on each motor.
- Narrow-frame test: the team built a 20 in x 32 in KoP frame with uncut end pieces so it could still go wider.
- Offseason swerve bring-up: the code set angle motor inversion before the factory reset. Three workarounds hid the bug until the team moved the inversion after the reset.
- Offseason joystick shaping: a polynomial curve on x and y separately locked driving to the cardinal directions; applying it to the polar radius fixed it.

Onshape CAD: 2023 robot CAD (Onshape) · 2023 robot CAD (Onshape) · 2023 Open Alliance build thread (CD) · Blender design model (Google Drive) · Offseason write-up in the next build thread (CD) · Offseason robot CAD (Onshape) · Offseason code (GitHub)

■ Study pick

555 Montclair Robotics

USA-NJ

Local EPA 34 (#987)

18-24

3-stage cascade elevator with horizontal scissor 'stinger', printed pneumatic gripper, roller-bar intake; swerve

SCORING (CONE / CUBE)

Pneumatic gripper: plates between two arms clasp the game piece. The gripper is 3D printed in carbon fiber filled high-temp PLA to save weight, because it sits on the end of the stinger, which the team calls 'a giant cantilever'. Standard PLA prototypes came first; the team moved to CAD and 3D printing after slow metal and polycarbonate prototypes in Week 2. In Week 6 the preferred plate surface was pool noodle covered with sandpaper. The January plan had free-spinning plates so a cone turns to the scoring orientation; the final plate design is unconfirmed (see CAD).

INTAKE

Changed in week 7. The original deployable wheeled intake (the 'schwooper') folded with pneumatic cylinders. Its metal prototype used green AndyMark compliant wheels. In January a cone fed base first sometimes caught between the wheels; the team planned nylon blockers between them. In testing at team 4361 on 2023-02-25 the cylinders could not deploy the intake from their mounts, and the team removed it. The replacement is a roller bar with wheels like the 2022 intake, built in one week with no CAD; the plan added a sheet metal ramp that funnels pieces into a collection area for the gripper.

ARM / ELEVATOR

A 3-stage cascade elevator carries a horizontal scissor 'stinger'; the team chose this over an extending arm on 2023-01-29. The elevator was re-CADed three or four times to fix a bearing block issue, and the CAD was complete by 2023-02-19. The elevator is rigged with springs, rope and chain. Changed in week 8: new parts for the scissor actuation, which the team expected to extend and retract faster; the January plan used springs and a lead screw. The battery mounts in a vertical electronics stack that helps counterbalance the robot when extending.

DRIVETRAIN

Swerve. The team built an MK4i swerve drivetrain in the 2022 offseason and stated it would use swerve in 2023. A 2023-02-22 post refers to the drive modules and to two steel bars across the front that support the drivetrain. The frame perimeter is 27x30 in; it grew from 25x30 in after an inflated cube did not fit a 10 in opening. Competition module type: Unconfirmed (see CAD).

ENDGAME

Auto balancing on the Charge Station worked in code by the Week 6 post (2023-02-19). The match dock/engage approach is Unconfirmed (not stated in the public release).

AUTONOMOUS

Charge Station auto balancing ran by 2023-02-19. TBA lists the Autonomous Award at FMA District Warren Hills; the team's post says the award was 'for the pretzel'. The routines are not described in the posts (see code).

VISION

Limelight with AprilTags tracked in 3D space. The team shorted one Limelight and broke another during build season, then got PhotonVision working. It wrote an interface that swaps between Limelight and PhotonVision by changing a few variables. At the District Championship it used a Limelight lent by team 341 after it chipped its own lens.

PUBLIC CAD / DESIGN

2023 robot CAD (Onshape)

LESSONS FOR STUDENTS

- Compare two architectures on paper first: the team framed elevator-plus-stinger as Cartesian (x then y) and the extending arm as polar (r then theta), and named the weak point of each.
- Prototype with an actual game piece before frame lock: an inflated cube did not fit the 10 in opening, so the frame changed from 25x30 in to 27x30 in.
- Prototype every mechanism: the team says the folding intake was the one piece it did not prototype, and its cylinders could not deploy it from their mounts.
- A simple fallback can be built fast: after the folding intake failed, the team built a roller-bar intake in one week without CAD.
- Keep mass off the end of a cantilever: the gripper is 3D printed in carbon fiber filled PLA because it rides on the end of the stinger.

Onshape CAD: 2023 robot CAD (Onshape) · 2023 build thread (CD) · 2023 robot CAD (Onshape) · Code (GitHub) · Kickoff design document (Google Docs) · Intake failure design-meeting notes (Google Docs) · Mount Olive event report (Basecamp) · Warren Hills event report (Basecamp) · District Championship event report (Basecamp)

■ Study pick

3737 Roto Raptors BB Mantis

USA-NC

Local EPA 20 (#2221)

12-16-2

'BB Mantis': cord-extended arm with passive claw, through-robot handling, all motors on chassis; 3D-printed PETG swerve

SCORING (CONE / CUBE) Passive game piece claw that closes on contact (release post). Cord on a spool driven by a 24:1 3D-printed cycloid gearbox resets the claw. Cone and cube handling detail: Unconfirmed (see CAD).	**INTAKE** Unconfirmed (see CAD). The release lists through-robot game piece manipulation; a separate floor or substation intake is not described.
ARM / ELEVATOR Arm driven by a NEO through a chain input stage, a 48:1 3D-printed cycloid gearbox and a chain output stage, for the 200:1 the team needed with a cone at full extension. A gas shock relieves load on the arm motors, and a compliant arm joint withstands side impact. Arm extension uses cord on a 24:1 cycloid spool. The published CAD shows the later VEX spur-gear arm gearbox.	**DRIVETRAIN** 3D-printed PETG swerve modules bolted under a 3030 T-slot chassis (swerve release). The released base design uses NEO drive and NEO 550 steer motors with a 4 in x 1.5 in wheel. The robot ran the team's TPU suspension tread wheels; the tread post states its 89 lb robot used 40% infill. All motors sit on the chassis for a low CG.
ENDGAME Charge Station: the team states the tread pattern was designed for grip on the Charge Station and let the robot climb it with ease. Dock and engage method: Unconfirmed (see code).	**AUTONOMOUS** Unconfirmed (see LabVIEW code).
VISION Unconfirmed (not stated in the public release)	**PUBLIC CAD / DESIGN** CAD and LabVIEW code (Google Drive)

LESSONS FOR STUDENTS

- A cycloid with three discs at 120 deg offset keeps nearly full tooth engagement and runs balanced; the team states a PETG-printed version carries more torque than expected.
- Upgrading one spur gear can move the failure: a steel 18T penultimate gear moved the failure to the final aluminum gear.
- Suspension tread infill sets the contact patch: the team aims for about 1 in of flat on the carpet and used 40% infill at about 90 lb and 50% near max weight.
- A printed swerve module can be documented like a product: the release states size (5.35 x 7.02 x 9.11 in), weight (2.5 lbs without motors, gearboxes and controllers), five drive ratios and a full BOM.
- Choose materials and fastening for printed parts: the team prints all parts except treads in PETG, avoids carbon-fiber filament for health reasons, and uses self-tapped threads with no brass inserts. Its 2026 posts give 4.5 mm holes for #10-32 screws, about 3x D deep for pull-out loads.

Drive folder / doc: CAD and LabVIEW code (Google Drive) · CAD and code release (CD) · CAD and LabVIEW code (Google Drive) · 3D-printed swerve release (CD) · Swerve CAD, BOM and instructions (Google Drive) · Suspension tread release (CD) · Suspension tread CAD (Google Drive)

■ Study pick

9062 Critical Circuits Bryce

Canada

Local EPA 19 (#2368)
15-34

'Bryce': long-range dual-flywheel cube shooter, two side roller intakes, 38 x 22 in MK4i L2 swerve

SCORING (CONE / CUBE) Dual flywheel cube shooter: 4 in Colson wheels on 2x NEOs at 1:1, inclined 15 degrees. The team reports a 35 ft range that proved inaccurate at that distance. A 1-button launch sequence runs the sequencer and the shooter flywheel assemblies. Low scoring works by outtaking through the side intakes.	**INTAKE** Two side roller intakes for cubes, each 22 in wide. A pneumatic cone 'nose' gripper, similar to 2019 hatch grabbers, was removed before DCMP due to limited use and effectiveness.
ARM / ELEVATOR No arm or elevator for scoring is described. The team lists a 'stick arm' and states a 6.5 in center of gravity regardless of stick arm location. A sequencer feeds the shooter.	**DRIVETRAIN** Swerve on MK4i modules at the L2 ratio. 38 in by 22 in chassis with a 1/8 in steel belly pan; the team states it is the longest swerve to its knowledge.
ENDGAME Charge Station balance is part of the 2 piece autonomous mode. Endgame dock or engage approach unconfirmed (not stated in the public release).	**AUTONOMOUS** A 2 piece balanced autonomous mode plus 4 alternative autonomous modes.
VISION Limelight 2+ with auto-alignment for cube node AprilTags.	**PUBLIC CAD / DESIGN** Onshape CAD (the team calls it incomplete)

LESSONS FOR STUDENTS

- A rookie team can specialize: the robot was designed for long-range cube scoring and tiebreaking defense, not cone placement.
- Report range and accuracy separately: the shooter reached 35 ft, but the team says it was inaccurate at that distance.
- Remove what does not earn its place: the pneumatic cone gripper and its full pneumatics system came off before DCMP.
- The team lists a long, narrow chassis (38 x 22 in) with a 1/8 in steel belly pan and a 6.5 in center of gravity regardless of stick arm location.
- Compare with the offseason robot 'Bill' (MK4i L3 on a 24 in square chassis, linear actuator shoulder, 180:1 elbow on 2x NEOs) to see how the second design changed.

Onshape CAD: Onshape CAD (the team calls it incomplete) · Robot showcase and season overview (CD) · 'Bryce' CAD (Onshape) · Offseason robot 'Bill' thread (CD) · 'Bill' CAD (Onshape) · 'Bill' code (GitHub)

Profile

663 Power Sharks MAKO

USA-MA

Local EPA 18 (#2546)
11-12

'MAKO': 6 NEO 6WD; electric chain-driven elevator, pneumatic arm extender, pneumatic claw on a motorized wrist

SCORING (CONE / CUBE) Pneumatic claw on a motorized wrist (the team's release post). Grip surface and wrist motor are unconfirmed (see CAD).	**INTAKE** Unconfirmed (see CAD). The release post names no separate intake; the claw is the only game piece mechanism it names.
ARM / ELEVATOR Electric chain-driven elevator with a pneumatic arm extender (the team's release post). Stage count and elevator motor are unconfirmed (see CAD).	**DRIVETRAIN** 6WD driven by 6 NEO motors (the team's release post). Wheel type and gearing are unconfirmed (see CAD).
ENDGAME Charge Station approach unconfirmed (not stated in the public release).	**AUTONOMOUS** Unconfirmed (not stated in the public release).
VISION Unconfirmed (not stated in the public release).	**PUBLIC CAD / DESIGN** Onshape CAD

LESSONS FOR STUDENTS

- The superstructure splits motion between an electric chain-driven elevator and a pneumatic arm extender. Compare where each motion type is used.
- The claw is pneumatic but the wrist under it is motorized. Study how the CAD routes air and power to the end of the arm.
- Drive base: 6 NEO motors on a 6WD. Check the gearbox layout in the CAD.

Onshape CAD: Onshape CAD · CAD release thread (CD) · Onshape CAD

Sources: the Spectrum 3847 CAD Collection, team CAD, binder and code releases, Chief Delphi release and reveal threads, and match data from The Blue Alliance. EPA is Local EPA (approx), computed independently. It is not official Statbotics data. "Unconfirmed" means the public material does not state the detail.